

OpenStax Chemistry 2e Errata

<https://openstax.org/details/books/chemistry-2e>

May 12, 2026 – Dean Peterson

Notes:

- I am a chemistry learner, not an expert, doing this avocationally not in a class setting.
- Since I lack instructor access, I could not review even numbered exercise solutions.
- I have attempted to include screenshots of unlabeled figures to assist in locating them.
- This document should have a PDF table of contents created from the headings.

Entire Book

Although this document mostly contains errata specifically for Chapter 7, this section contains some issues which also apply more broadly.

Accessibility of Detailed Element Information

There are multiple periodic table shaped diagrams throughout the book. Some of these diagrams include the details of every cell in the periodic table in the alt text, while other diagrams omit the details from the alt text, providing no accessible source of those details.

I am not an accessibility expert. However, would it be more accessible to include one or more HTML tables of element properties in the book? They could be located in Appendix A or in their own Appendix. They would not have to be formatted as periodic tables. They could just be formatted as normal data tables with one row per element. Putting the information in HTML tables should allow the use of whatever assistive technologies already exist for tabular data. Perhaps the alt text for the various periodic table shaped diagrams could refer to that appendix.

Here is a list of periodic table shaped diagrams that I have found in just the first 7 chapters, most of which might benefit from this treatment. The **Alt Text Status** column represents whether the element details are present in the alt text or not.

Figure	Description	Element Information	Alt Text Status	Needed For
2.26, and A1 in Appendix A	Periodic table	Symbol Atomic mass Name Period Group Metal/Nonmetal/Metalloid Solid/Liquid/Gas	Details included	?

Figure	Description	Element Information	Alt Text Status	Needed For
1.22	Periodic table	Symbol Atomic mass Name Period Group Metal/Nonmetal/Metalloid Solid/Liquid/Gas	Details omitted	?
2.27	Groups	Group names	Details included	?
2.29	Common ionic charges	Common ions	Details included	?
6.27	Outer subshells	Last added subshell Electron count in subshell	Electron count omitted	?
6.29	Outer electron configuration	Outer shell electron configuration	Details omitted	?
6.30	Covalent radii	(Pictures of sizes, so n/a?)	Size trends included	?
6.34	First ionization energies	First ionization energy	Details included	?
6.35	Electron affinities	Electron affinity	Details included	?
7.6	Electronegativity	Pauling electronegativity	Details omitted	Example 7.3

Accessibility of Other Figures Containing Tables

Chapters 7 and 8 contain multiple figures which are basically tables with images in some of the cells. At least a couple of these figures are quite large, leading to enormous alt text. Would it be more accessible to make these figures real HTML tables? Putting the information in HTML tables should allow the use of whatever assistive technologies already exist for tabular data. Cells with graphic images would still need alt text, but at least that alt text would be broken up into separately accessible pieces for each cell that might be easier to navigate.

Tables this might be helpful for include:

- Figure 7.9
- Figure 7.10
- Figure 7.16 (quite large)
- Figure 7.19 (quite large)
- Figure 8.21

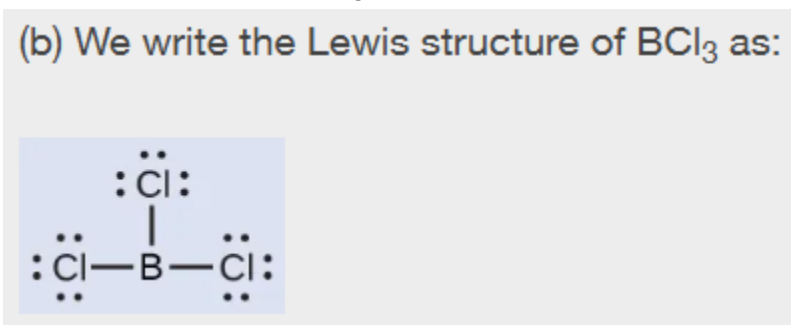
Inconsistent Formatting of Unlabeled Figures

Most unlabeled figures appear on their own line. Put another way, they appear to be their own block level element. However, in the web version of the book, some unlabeled figures appear on the same line as the preceding text. (The PDF version seems to show all unlabeled figures on their own line.)

As examples, consider the first two unlabeled figures in **Chapter 7.6, Example 7.11, Solution section**. The unlabeled figure under point (a) appears on the same line. This was tested on Firefox and Chrome on Windows 11.



In contrast, the unlabeled figure under point (b) appears on its own line.



Another set of examples are the two unlabeled figures in **Chapter 8.2, Example 8.2**, the first of which appears on the same line as the preceding line in the web version. The second appears on its own line. These are only examples, there are probably other unlabeled figures with this issue.

Looking at the underlying HTML, it appears the difference is not the HTML for the unlabeled figure itself. Rather, it appears that the difference is whether the preceding text appears in its own paragraph <p> element or not.

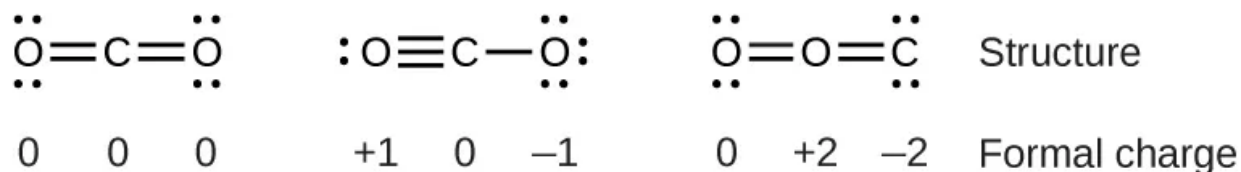
Note: End of line wrapping can make an unlabeled figure that is on the same line as the preceding text appear to be on its own line. I used different browser widths to double check that this is not the case with the examples cited above.

Notation Variation Used Without Being Introduced

I think Lewis structures with the formal charges only labeled on some atoms is never really introduced before it is used. Furthermore, using + and - instead of +1 and -1 labels on atoms

makes it look even more different. I am still only 95% sure that the Lewis structures with charge indicators on only some atoms are representing formal charges even now.

Chapter 7.4, section Using Formal Charge to Predict Molecular Structure, implicitly introduces Lewis structures that have the atoms labeled with their formal charges. The examples show Lewis structures with all atoms labeled with either “0” or a sign and a number “-1”, “+1”, “-2”, etc., like the screenshot below.



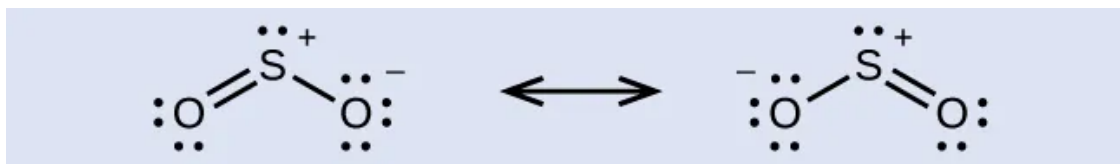
In contrast, at a few places in Chapters 7 and 8, Lewis structures are shown with what I assume are formal charges next to atoms, but shown as “+” instead of “+1”, “-” instead of “-1”, and leaving atoms with zero formal charge unlabeled. Note that these are cases where the charge indicators are next to the atoms, not outside of brackets such as a charge for a whole ion.

This confused me as a learner, because I did not know (and still am not 100% certain) that the latter notation represents formal charges. Furthermore, some exercise answers show the +/- nomenclature, which was never officially introduced, and which was not asked for in the exercise questions. I have created separate errata for some of these. However, the best solution might be to introduce that alternate or variant notation in the book text. Here is list of the places I know the +/- nomenclature appears:

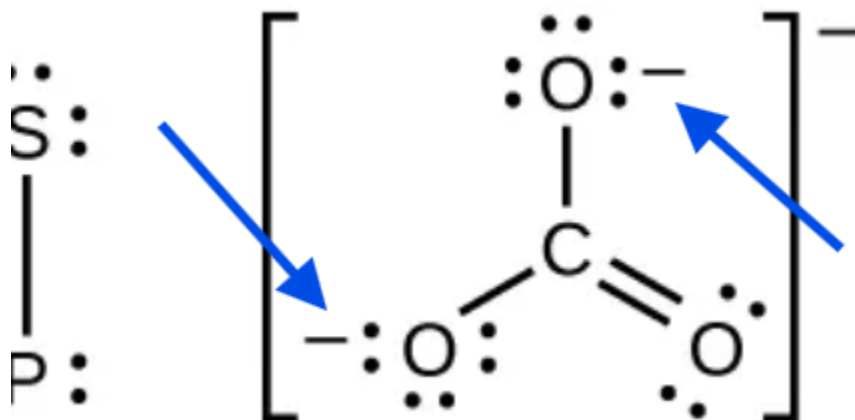
- Exercise 7.29 (k), solution unlabeled image



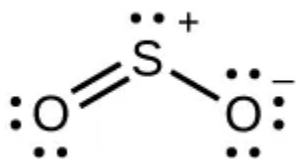
- Chapter 8.3, Example 8.4, Solution section, unlabeled image



- Exercise 8.19 (a), solution unlabeled image, right half showing ClO_3^- ion.



- Exercise 8.30 (b), question unlabeled image



Chapter 7

Pyramid Terminology Small Inconsistency

Throughout the chapter, sometimes “bipyramid” is used, and sometimes “bipyramidal” is used, often in identical situations. This may include the alt text, exercises, and solutions. It is not a big deal, but it might be worth searching for and changing whichever form is not preferred. To give just one example, Exercise 7.91 (b) solution has “trigonal bipyramid”, while Exercise 7.95 (f) solution, “trigonal bipyramidal”. Some instances of the terms appear in figures such as Figure 7.16 and Figure 7.19 as well. There is also a single example of “bypyramidal”, with two “y”s, in the Exercise 7.93 (e) solution. A separate errata exists for that misspelling later in this document.

Molecular Simulator Accessibility

Chapter 7 uses molecular simulators. They are not only linked as an additional reference, but are required for some examples and exercises. This makes it critical that these resources be accessible. I was unable to use keyboard shortcuts on the simulations referenced in this chapter, which makes me wonder if they are accessible.

Here some examples and exercises that require the molecular simulators:

- Example 7.17, all sections
- Example 7.18, all sections
- Exercises 7.112 through 7.116

Notes:

- Some or all of the simulators used in this chapter are on the [pan and zoom accessibility list](#) at PhET. However, I was not able to discover keyboard shortcuts to enter the simulations from menus or change simulation options. I could only pan and zoom.
- I tested for keyboard shortcuts on Firefox 149.0.2 (64-bit) on Windows 11.
- I do not have such challenges, so I do not have accessibility tools installed, so I may be missing something.
- It is possible that the old Java-based versions and the newer HTML5-based versions have different levels of accessibility support.
- No HTML5-based version appears to be available for the “real molecule” section of the molecular polarity simulation used in Exercise 7.113. (There is a separate errata on that issue in this document.)

Lewis Structure red dots

Some Lewis structures in this chapter use red dots to show where certain electrons go. As someone who is partially red/green colorblind, it is a little hard to distinguish between the red dots and the black dots. However, I do not think this hampered my understanding in this case.

Chapter 7.1

Figure 7.1 alt text

Sentence 2 reads (emphases mine):

The left figure is a **many-sides** spherical ball **composed of hexagonal rings** which have carbon atoms at each corner.

Should “many-sides” be “**many-sided**” with a final “d”? Also, I think “composed of hexagonal rings” should be “**composed of hexagonal and pentagonal rings**”?

Sentence 4 reads:

The right picture shown as water tower with sides shaped like hexagonal rings. Shouldn’t “shown as” be “shows a”. Also, I think “shaped like hexagonal rings” should be “shaped like hexagonal and pentagonal rings”?

Figure 7.3 alt text issues

The final sentence reads (emphases mine):

The purple spheres are labeled “**N** superscript **postive** sign” while the green are labeled “**C** superscript negative sign.”

The “N” should be “**N a**”, and the word “**positive**” is misspelled “postive”.

Figure 7.3 alt text suggestion

Saying that purple and green spheres are alternating does not fully capture the geometry. What about adding a final couple of sentences.

In both diagrams, the geometry is cubic, with each ion having six neighboring ions, all of which are the other type of ion. In the right diagram, all of the white rods linking the ions are at right angles to each other.

Ch 7.1, section Electronic Structures of Cations, paragraph 4 idea

Background:

When I got to this section, and even before looking at the electron configurations in exercises 6.57 and 6.65, I was surprised to see electron configurations given in orders other than the aufbau energy order given in Chapter 6.4. For example, I was expecting $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ instead of $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$, and $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1 4p^6$ instead of $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2 4p^6$. I eventually figured out that ordering by shell is more useful for ions where s subshell electrons are lost first.

Idea:

Would it make sense to mention somewhere that giving electron configurations in an alternate order is acceptable as long as all subshells are present with the appropriate counts. Perhaps it could go in the Chapter 7.1 section Electronic Structures of Cations, paragraph 4, where expressing electron orbitals in such an alternate order seems to become preferable.

Chapter 7.2

Figure 7.4 alt text

Sentence 2 ends (emphasis mine):

... two values: “0” at the far left and “**0.74**” to the left, are labeled on the x-axis.

The “0.74” should be “**74**”.

Figure 7.5 alt text

The text currently reads:

Two diagrams are shown and labeled “a” and “b.” Diagram a shows a small sphere labeled, “H” and a larger sphere labeled, “C I” that overlap slightly. Both spheres have a small dot in the center. Diagram b shows an H bonded to a C I with a single bond. A dipole and a positive sign are written above the H and a dipole and negative sign are written above the C I. An arrow points toward the C I with a plus sign on the end furthest from the arrow's head near the H.

There are three issues with this text.

- The final sentence is about the label on diagram a but is positioned at the end after the alt text has switched to describing diagram b.
- For diagram (a), no mention is made of the shading, which I believe indicates electron probability density. There should be a mention that the left atom is shaded with a light green which blends into darker shading in the right atom.
- The term “dipole” is used in the alt text but not used in the main text. The main text just mentions using the Greek letter delta, not the term “dipole”.

Suggested replacement alt text (changes in bold):

Two diagrams are shown and labeled “a” and “b.” Diagram a shows a small sphere labeled, “H” and a larger sphere labeled, “C I” that overlap slightly. Both spheres have a small dot in the center. **Both spheres are shaded green, but the green shading gradually becomes darker as you move from the H sphere to the C I sphere. An arrow above the diagram points from the H side toward the C I side, with a plus sign on the end furthest from the arrow's head near the H.** Diagram b shows an H bonded to a C I with a single bond **shown as a short single line. Above the H symbol is a lowercase Greek letter delta followed by a positive sign. Above the C I symbol is a lowercase delta followed by a negative sign.**

Figure 7.6

Is there a reason that As and At are colored as metalloids in the periodic table in Appendix A, but not colored as metalloids in the Pauling electronegativity table in Figure 7.6. Note that the colors are not even labelled in Figure 7.6, so it may not matter.

Figure 7.8 alt text

Sentence 1 reads:

Two flow charts and table are shown.

Shouldn't it be?

Two flow charts and **a** table are shown.

Sentence 6 starts:

A double ended arrow is written vertically to the right of the flow charts ...

The arrow is not double ended. It is downward-facing, with an arrowhead only on the bottom.

Chapter 7.3

Unlabeled figure of Ca Lewis symbol, alt text

This figure is in chapter 7.3, section Lewis Symbols, just after the first paragraph.

The current alt text reads:

A Lewis structure of calcium is shown. A lone pair of electrons are shown to the right of the symbol.

Some issues:

- It says the 2 electrons are paired to the right of the symbol. They are not. One is on the left, one on the right, and neither is paired.
- It does not explicitly say what symbol is at the center, just calling it “symbol”.
- It does not describe that the electrons in the Lewis symbol are represented by black dots. This is the first Lewis symbol shown, so I think it would be wise to be more descriptive just this once.
- It calls the image a “Lewis structure”. Isn’t it technically a “Lewis symbol”, not a structure? Doesn’t a Lewis structure involve bond? That is what the later section on Lewis Structures says: “... Lewis structures, drawings that describe the bonding in molecules and polyatomic ions.”

Suggested replacement alt text:

A Lewis symbol of calcium is shown. In the center is the two-letter symbol for calcium, “Ca”. Two black dots representing electrons are shown, one along the left side of the “Ca,” and one along the right side of the “Ca.”

Figure 7.9 alt text

The sentence 4 description of the second column starts as follows:

The second column contains the symbols and numbers “[Ne] 3 s superscript 2,” “[Ne] 3 s superscript 2, 3 p superscript 1,” ...

The first non-header row is missing from that description. It should start (insertion in bold):

The second column contains the symbols and numbers “[**Ne] 3 s superscript 1,” “[Ne] 3 s superscript 2,” “[Ne] 3 s superscript 2, 3 p superscript 1,” ...**

2nd unlabelled figure after Figure 7.9 alt text

Sentence 2 starts:

The left diagram shows a Lewis dot structure of chlorine with seven dots and the letter “e” with a superscripted negative sign, ...

This does not include the + sign between Ca and e⁻. The alt text for the similar figure just above it does include the + sign. Suggest adding it:

The left diagram shows a Lewis dot structure of chlorine with seven dots, **a plus sign**, and the letter “e” with a superscripted negative sign, ...

Sentence 4 has the same issue. Suggest changing it to start:

The right diagram shows a Lewis dot structure of sulfur with six dots, **a plus sign**, and the symbol “2e” with a superscripted negative sign, ...

Figure 7.10 alt text

Sentence 4 starts:

A sodium symbol with one dot, ...

It is not mentioned that the Na dot is red, which is mentioned for the Mg and Ca dots later. Suggest mentioning it here too so they know where the red dot in sentence 6 came from.

A sodium symbol with one **red** dot, ...

Sentence 10 ends (emphasis mine):

... two of which are red, surrounded by brackets with a superscripted two **a** and a negative sign.

The bold word “a” should be removed.

2nd unlabelled figure in chapter 7.3 section Lewis Structures, alt text



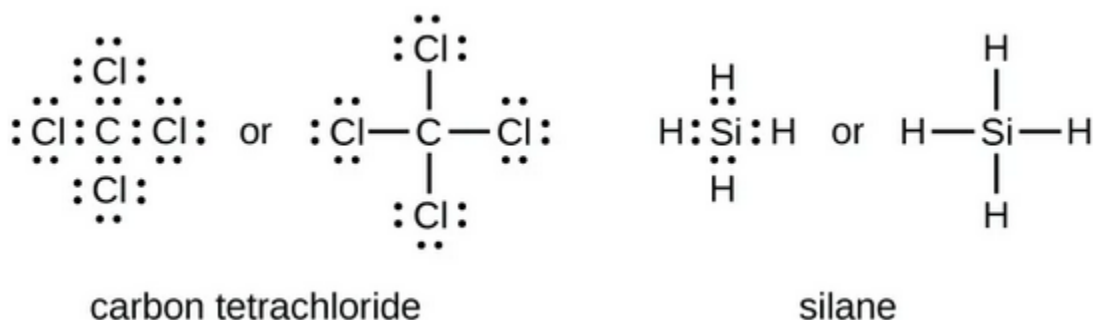
Sentence 2 reads:

The left-hand structure shows two H atoms connected by a single bond.

I believe this is the first time a line-shaped bond indicator is used with a Lewis structure, so being a bit more descriptive could be in order. I suggest inserting a new sentence 3 after sentence 2 like this:

The bond is indicated with a short solid line.

1st unlabelled figure in section The Octet Rule, alt text



The alt text reads:

Two sets of Lewis dot structures are shown. The left structures depict five C l symbols in a cross shape with eight dots around each, the word “or” and the same five C l symbols, connected by four single bonds in a cross shape. The name “Carbon tetrachloride” is written below the structure. The right hand structures show a S i symbol, surrounded by eight dots and four H symbols in a cross shape. The word “or” separates this from an S i symbol with four single bonds connecting the four H symbols in a cross shape. The name “Silane” is written below these diagrams.

There is one mistake and a couple of things are ambiguous:

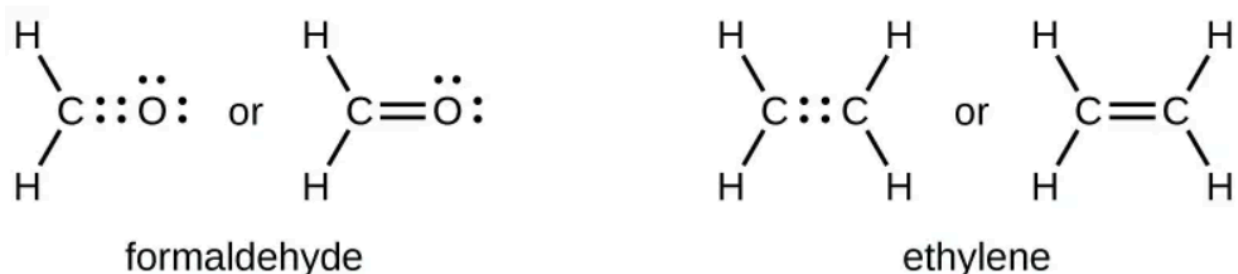
- It is not 5 chlorine atoms, it is 1 carbon atom surrounded by 4 chlorine atoms.
- Sentence 2 says there are eight dots around each atom. This could be interpreted as each atom having its own full independent set of eight dots rather than sharing the dots between them. (This seems especially relevant since the next section is on double bonds, where there are two pairs of dots between atoms and that is what distinguishes it from a single bond.)
- Sentence 2 does not make it explicit that there are no lines for bonds in the left-most diagram.
- Sentence 2 does not describe what the “single bonds” look like in the figure after the first “or”. Technically, the diagrams on both sides of the “or” show atoms “connected by four single bonds.” It is only how those bonds are depicted that differs. I suggest describing the single bonds after the “or” as being depicted by short, solid lines.
- In sentence 2, no mention is made of the dots still present on the Cl atoms after the “or”, leaving it a bit ambiguous which dots were replaced by the line segment single bond notation.

Here is my attempt at a revision.

Two sets of Lewis dot structures are shown. The leftmost structure depicts a central C symbol surrounded by four C l symbols in a cross shape. There are eight dots around each atom, but the pairs of dots between atoms are shared, such that there are only two dots between the C atom and any C l atom. No bonds are depicted with lines in this structure. The leftmost structure is followed by the word “or” and a similar structure where the pair of dots between each C l atom and the C atom are replaced with a short line segment representing the single bond. The other six dots around each C l remain. The name “Carbon tetrachloride” is written below the first two structures. The third structure shows a S i symbol, surrounded by eight dots and four H symbols in a cross

shape. The word “or” separates this from a similar structure where the pair of dots between each H atom and the S i atom are replaced with a short line segment representing the single bond. With this replacement, there are no dots remaining in this right most structure. The name “Silane” is written below these last two structures.

1st unlabelled figure in section Double and Triple Bonds, alt text



Sentences 3 and 4 read:

There are four electrons between the C atom and an O atom. The O atom also has two pairs of dots.

Referring to the four dots between the C atom and an O atom as “electrons” but the lone pairs on the oxygen atom “dots” could give the impression that they are different things. What about this? (change in bold)

There are four **electron** dots between the C atom and an O atom. The O atom also has two lone pairs of dots.

Sentence 5 reads:

The word “or” separates this structure from the same diagram, except this time there is a double bond between the C atom and O atom.

Referring to the double lines as “a double bond” while previous sentences mention “electrons” or “dots” obfuscates the fact that both diagrams show a double bond, they just show it in different ways. What about this?

The word “or” separates this structure from the same diagram, except this time the double bond between the C atom and O atom is indicated with a double line in place of the 4 dots.

Sentence 9 reads:

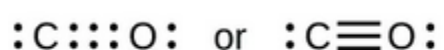
The word “or” lies to the left of the second structure, which is the same except that the C atoms form double bonds with one another.

This has the same issue as sentence 5 above. Also, the previous focus was on the third diagram, so why does it start by saying the word “or” is to the left of the coming structure, a structure just being introduced? I suggest reusing sentence 5’s phrasing for the location of the word “or”.

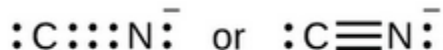
Here is a suggested replacement sentence 9 that addresses those issues:

The word “or” separates this structure from a similar structure where the double bond between the C atoms is indicated with a double line in place of the 4 dots.

2nd unlabelled figure in section Double and Triple Bonds, alt text



carbon monoxide



cyanide ion

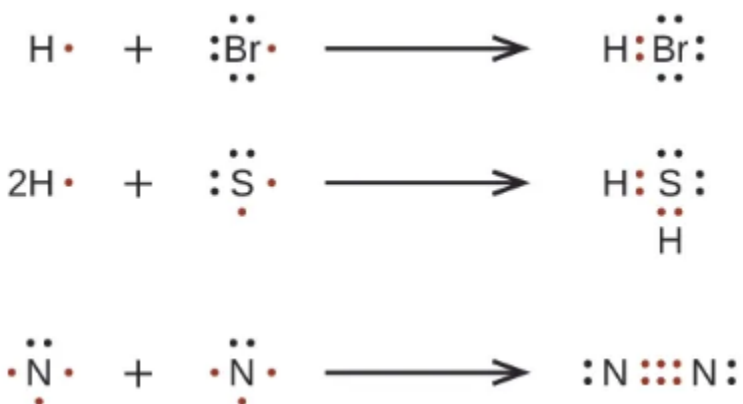
The alt text reads:

Two pairs of Lewis structures are shown. The left pair of structures show a C atom and an O atom with six dots in between them and a lone pair on each. The word “or” and the same structure with a triple bond in between the C atom and O atom also are shown. The name “Carbon monoxide” is written below this structure. The right pair of structures show a C atom and an N atom with six dots in between them and a lone pair on each. The word “or” and the same structure with a triple bond in between the C atom and N atom also are shown. The name “Cyanide ion” is written below this structure.

This is missing the minus signs on the second set of diagrams. It also has the same issue as above. By saying that a “triple bond” replaces “six dots” it implies that the “six dots” did not already represent a triple bond. This also gives no indication as to the visual line symbol used for a triple bond. How about the following (changes in bold):

Two pairs of Lewis structures are shown. The left pair of structures show a C atom and an O atom with six dots in between them and **another lone pair of dots on each atom**. The word “or” and the same structure **follow, but with the triple bond between the C atom and the O atom indicated by a triple line in place of the six dots**. The name “Carbon monoxide” is written below this structure. The right pair of structures show a C atom and an N atom with six dots in between them and **another lone pair of dots on each atom**. **There is a superscripted negative sign after the N atom**. The word “or” and the same structure **follow, but with the triple bond indicated with a triple line in place of the six dots**. The name “Cyanide ion” is written below this structure.

1st unlabeled figure in section Writing Lewis Structures with the Octet Rule, alt text



Sentence 5 reads:

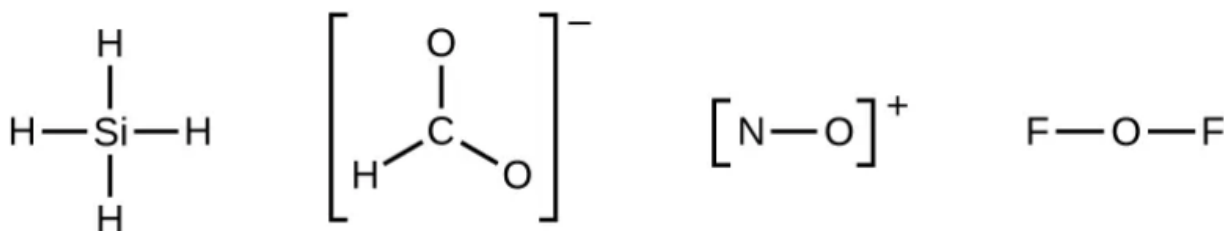
There are two red dots in between the two hydrogen atoms and the sulfur atom.

I suggest the following clarification (change in bold):

There are two red dots in between **each of** the two hydrogen atoms and the sulfur atom.

This removes any ambiguity between their being two and four red dots total while also helping the reader know that the sulfur atom is in the middle of the three, not one of the hydrogen atoms.

2nd unlabeled figure in section Writing Lewis Structures with the Octet Rule, alt text



Sentence 2 reads (emphasis mine):

The first shows one silicon single **boned** to four hydrogen atoms.

The word “boned” should be “**bonded**”.

Sentence 3 reads (emphasis mine):

The second shows a carbon which forms a single bond with an oxygen and a hydrogen and a **double** bond with a second oxygen.

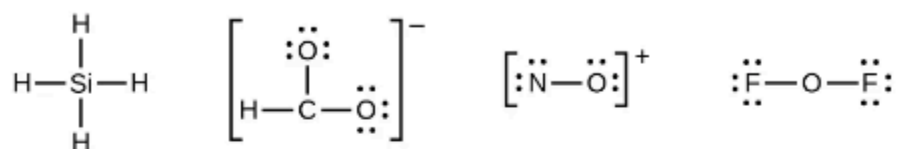
Neither of the oxygens are joined to the carbon with a double bond. Suggest:

The second shows a carbon which forms single bonds with a hydrogen and two oxygen.

3rd, 4th, and 5th unlabeled figures in section Writing Lewis Structures with the Octet Rule, formatting

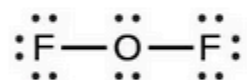
3. Distribute the remaining electrons as lone pairs on the terminal atoms to give them their valence shells with an octet of electrons.

- There are no remaining electrons on SiH_4 , so it is unchanged.



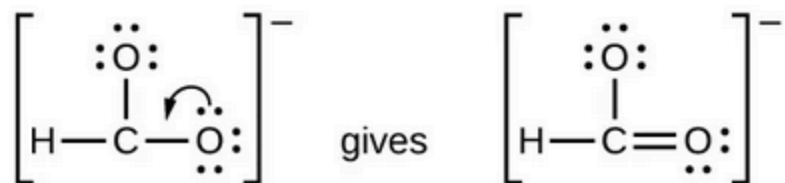
4. Place all remaining electrons on the central atom.

- For SiH_4 , CHO_2^{-} , and NO^{+} , there are no remaining electrons determined in Step 1.
- For OF_2 , we had 16 electrons remaining in Step 3, and we place them on the central atom:



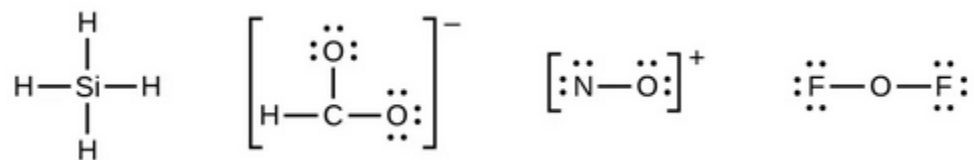
5. Rearrange the electrons of the outer atoms to make multiple bonds to obtain octets wherever possible.

- SiH_4 : Si already has an octet, so nothing needs to be done.
- CHO_2^{-} : We have distributed the valence electrons as shown, but the carbon atom lacks an octet:



These images are much smaller than the others, at least on the web. It makes them a little harder to read.

3rd unlabeled figure in section Writing Lewis Structures with the Octet Rule,
alt text



The 2nd sentence reads (emphasis mine):

The first shows one silicon single **boned** to four hydrogen atoms.

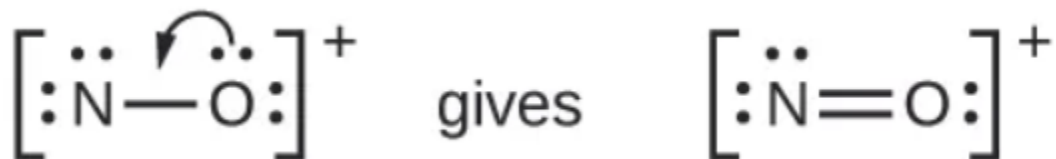
The word “boned” should be “**bonded**”

The 5th sentence reads (emphasis mine):

The third structure shows a nitrogen single bonded to an oxygen, each with **three** lone pairs of electrons.

Each only have **two** lone pairs of electrons, not three.

6th unlabeled figure in section Writing Lewis Structures with the Octet Rule,
alt text

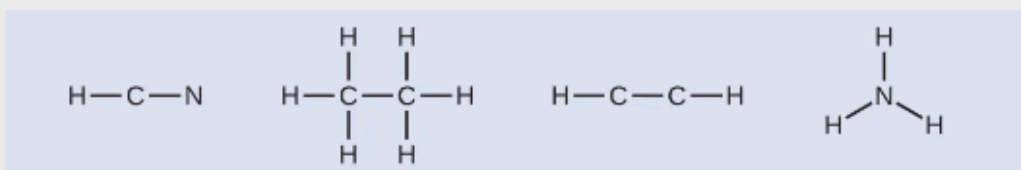


Unlike the alt text on the figure before it, this alt text omits describing the curved arrow. I suggest inserting a new sentence 3:

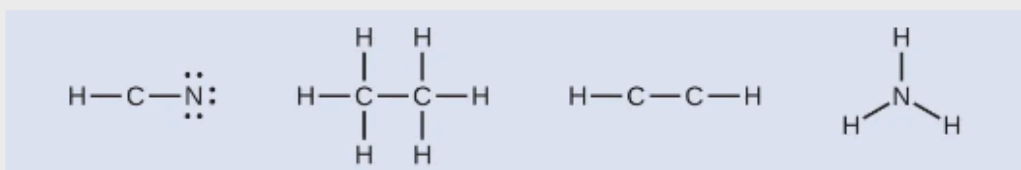
A curved arrow points from one lone pair of dos on the oxygen atom to the single bond with the nitrogen atom.

Example 7.4 Solution section, first 3 unlabeled images, formatting

Step 2. *Draw a skeleton and connect the atoms with single bonds. Remember central atom:*



Step 3. *Where needed, distribute electrons to the terminal atoms:*



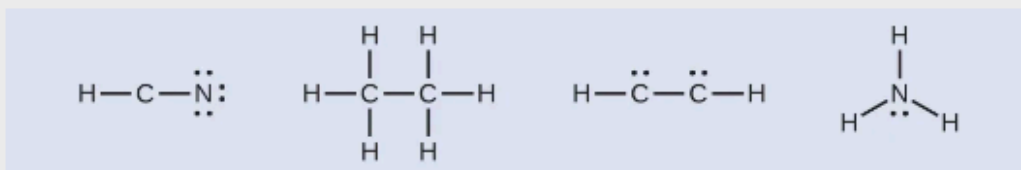
HCN: six electrons placed on N

H₃CCH₃: no electrons remain

HCCH: no terminal atoms capable of accepting electrons

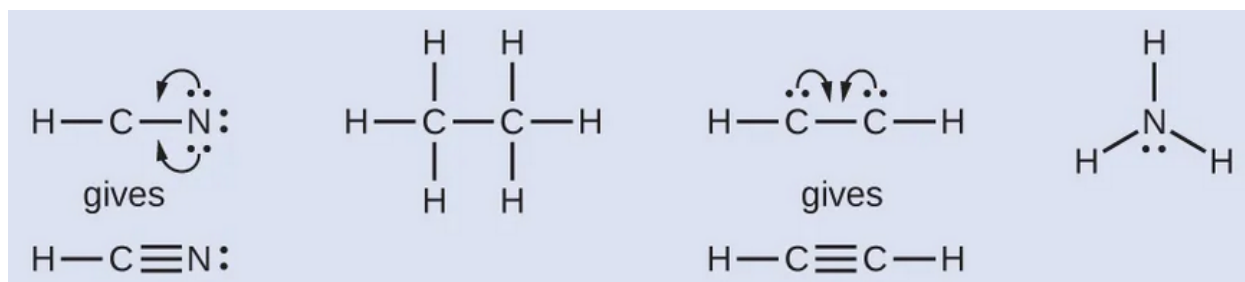
NH₃: no terminal atoms capable of accepting electrons

Step 4. *Where needed, place remaining electrons on the central atom:*



Again there is a size mismatch. The first 3 images in Example 7.4 are a lot smaller than the fourth image.

Example 7.4, final unlabeled image in Solution section, alt text



Sentences 3 and 4 read:

Two curved arrows point from the nitrogen to the carbon. Below this structure is the word “gives” and below that is the same structure, but this time there is a triple bond between the carbon and nitrogen.

Sentence 4 is missing the other difference between the top and bottom structures, which is the fact that the nitrogen now has one lone pair of electrons, not three. I suggest (change in bold):

Two curved arrows point from the nitrogen to the carbon. Below this structure is the word “gives” and below that is the same structure, but this time there is a triple bond between the carbon and nitrogen **and the nitrogen only has one lone pair of electrons, not three.**

Sentences 7 and 8 read:

Two curved arrows point from the carbon atoms to the space in between the two. Below this structure is the word “gives” and the same structure, but this time with a triple bond between the two carbons.

This is a similar issue as with sentence 4, where sentence 8 omits the other change from top to bottom, the removal of the lone pairs on the carbon atoms. I suggest (change in bold):

Two curved arrows point from the carbon atoms to the space in between the two. Below this structure is the word “gives” and the same structure, but this time with a triple bond between the two carbons, **each of which now has no lone pairs of electrons.**

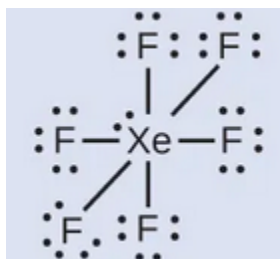
Box How Sciences Interconnect, alt text

Paragraph 1, sentence 1 ends:

... since prehistoric times, .

The comma and space after the word “times” but before the period should be removed.

Example 7.5, Solution section, 3rd unlabeled figure, alt text



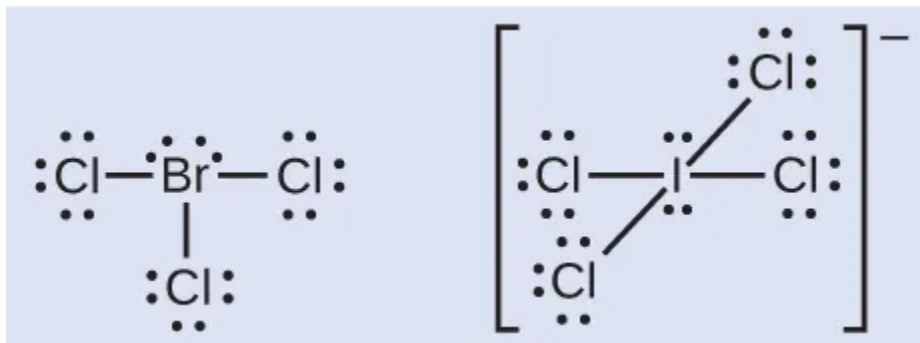
The alt text reads:

This structure shows a xenon atom single bonded to six fluorine atoms. Each fluorine atom has three lone pairs of electrons.

This forgets about the single lone pair of electrons on the xenon atom. Suggest adding a new final sentence:

The xenon atom has one lone pair of electrons.

Example 7.5, Check Your Learning section unlabeled figure, alt text



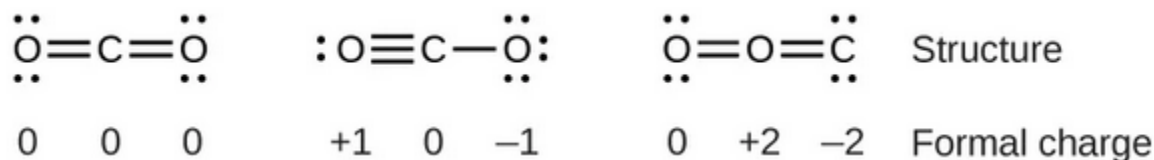
Sentence 3 reads (emphasis mine):

The right shows an iodine atom, with two lone pairs of electrons, single **boned** to four chlorine atoms, each with three lone pairs of electrons.

The word “boned” should be “**bonded**”.

Chapter 7.4

1st unlabeled figure, section Using Formal Charge to Predict Molecular Structure, alt text



The alt text currently reads (emphases mine):

Three Lewis structures are shown. The **left and right structures show** a carbon atom double bonded to two oxygen atoms, each of which has two lone pairs of electrons. The **center** structure shows a carbon atom that is triple bonded to an oxygen atom with one lone pair of electrons and single bonded to an oxygen atom with three lone pairs of electrons. The **third** structure shows an oxygen atom double bonded to another oxygen atom with **to** lone pairs of electrons. The first oxygen atom is also double bonded to a carbon atom with two lone pairs of electrons.

The alt text omits the formal charge numbers and has other problems detailed below, but first, a suggested replacement. This replacement is modeled after alt text for 2nd unlabeled figure in same section:

A row of three Lewis structures is shown with a row of numbers below it. The top row is labeled, "Structure" and the bottom row is labeled, "Formal charge." The left structure shows an oxygen atom with two lone electron pairs double bonded to a carbon atom that is double bonded to another oxygen atom with two lone electron pairs. Below this structure are the numbers zero, zero, and zero. The middle structure shows an oxygen atom with one lone electron pair triple bonded to a carbon atom that is single bonded to an oxygen atom with three lone electron pairs. Below the structure are the numbers positive one, zero, negative one. The right structure shows an oxygen atom with two lone electron pairs double bonded to another oxygen atom that is double bonded to a carbon atom with two lone electron pairs. Below the structure are the numbers zero, positive two, negative two.

The suggested replacement addresses the following issues. Things that are wrong:

- The formal charge numbers are never given.
- Sentence 2 begins "The left and right structures show ..." but proceeds to describe only one structure. "The left structure shows ..." would be better. The sentence 2 description cannot apply to the right structure as well because in the right structure the carbon atom is on the end and cannot be bonded to two oxygens as sentence 2 says.
- At the end of sentence 4, where it says "with to lone pairs of electrons", the word "to" should be "two".

Things that are inconsistent:

- In their respective sentences, the three structures are referred to as "left and right", "center", and "third", with the "third" not being parallel, even if the "left and right" is fixed. I suggest replacing "third" with "right" for consistency.

Things that are potentially confusing:

- In describing the third structure, sentence 4 discusses "an oxygen atom double bonded to another oxygen atom with to [sic] lone pairs of electrons". With that context, sentence

5 could cause confusion because it starts “The first oxygen atom ...” Does “first” refer to the first oxygen atom in the line of three atoms, which is the left atom, or the first oxygen atom discussed in the previous sentence, which is the center atom.

- The point of this section is the order in which these atoms should be connected. Given that, and given the fact that the reader must map atoms to formal charge numbers, it seems more intuitive to describe all structures from left to right, instead of starting with the center atom then jumping to the left and to the right.

2nd unlabeled figure section Using Formal Charge to Predict Molecular Structure, alt text

Structure	$[\ddot{\text{N}}=\text{C}=\ddot{\text{S}}:]^{-}$	$[\ddot{\text{C}}=\text{N}=\ddot{\text{S}}:]^{-}$	$[\ddot{\text{C}}=\text{S}=\ddot{\text{N}}:]^{-}$
Formal charge	-1 0 0	-2 +1 0	-2 +2 -1

The first part of the alt text, the part that describes the left structure, currently reads:

Two rows of structures and numbers are shown. The top row is labeled, “Structure” and depicts three Lewis structures and the bottom row is labeled, “Formal charge.” The left structure shows a carbon atom double bonded to a nitrogen atom with two lone electron pairs on one side and double bonded to a sulfur atom with two lone electron pairs on the other. The structure is surrounded by brackets and has a superscripted negative sign.

Below this structure are the numbers negative one, zero, and zero. ...

Here is a suggested replacement for just the first part of the alt text describing the left structure:

Two rows of structures and numbers are shown. The top row is labeled, “Structure” and depicts three Lewis structures and the bottom row is labeled, “Formal charge.” The left structure shows a nitrogen atom with two lone electron pairs double bonded to a carbon atom double bonded to a sulfur atom with two lone electron pairs. The structure is surrounded by brackets and has a superscripted negative sign. Below this structure are the numbers negative one, zero, and zero. ...

I suggest replacing the description of the left structure because:

- The atoms in the left structure are described in the order: center atom, left atom, right atom. Then the formal charge numbers are given in a different order, left atom, center atom, right atom. This is confusing.
- The atoms in the middle and right structures are described in the order: left atom, center atom, right atom, which maps better to the order the formal charges are given in.

I find the order used to describe the middle and right structures more clear, so suggest using it for the left structure as well.

1st unlabeled figure in section Resonance



Sentence 1 of the alt text reads:

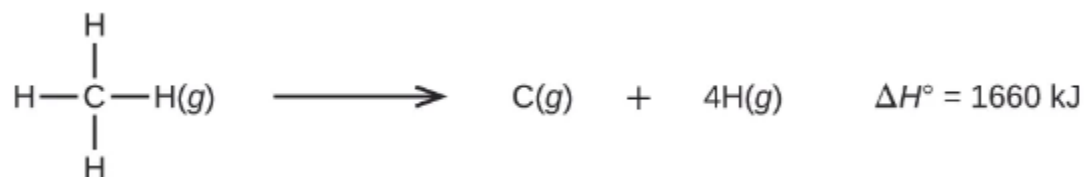
Two Lewis structures are shown.

To put the reader in the correct frame of mind before detailing the structures, I suggest expanding sentence 1 to read:

Two Lewis structures are shown which are essentially mirror images of each other, identical except for order.

Chapter 7.5

1st unlabeled image, section Bond Strength: Covalent Bonds



This is another image that seems unusually small, at least when viewed on the web.

Example 7.9

Paragraph 1, sentence 3 starts:

Using the bond energies in Table 7.3, calculate ...

However, the bond energies table, at least the one with all bonds required, is Table 7.2. The "Table 7.3" is a link, and both the text and the link destination are wrong.

Figure 7.13 alt text

The alt text reads:

A diagram is shown. An upward facing arrow is drawn to the far left of the chart and is labeled "H increasing." A horizontal line is drawn at the bottom of the chart. A downward-facing, vertical arrow to the left side of this line is labeled, "Overall change." Beside this arrow is another label, "capital delta H subscript f, equals negative 553.5 k J per mol, (Enthalpy of formation)." Three horizontal lines, one above the other, and all above the bottom line, are labeled, from bottom to top, as: "C s (s), plus sign, one half F

subscript 2, (g),” “C s (g), plus sign, one half F subscript 2, (g),” and “C s, superscript positive sign, (g), plus sign, one half F subscript 2, (g).” Each of these lines is connected by an upward-facing vertical arrow. Each arrow is labeled, “capital delta H subscript 1, equals 76.5 k J per mol, (Enthalpy of sublimation),” “capital delta H subscript 2, equals 375.7 k J per mol, (ionization energy),” and “capital delta H subscript 3 equals 79.4 k J / mol (one half dissociation energy).” Another horizontal line is drawn in the center top portion of the diagram and is labeled “C s, superscript positive sign, (g), plus sign, F, (g).” There is one more horizontal line drawn to the right of the overall diagram and located halfway down the image. An arrow connects the top line to this line and is labeled, “capital delta H equals negative 328.2 k J / mol (electron affinity).” The line is labeled, “C s superscript positive sign (g) plus F superscript negative sign (g).” The arrow connecting this line to the bottom line is labeled, “negative capital delta H subscript lattice equals negative 756.9 k J / mol.” The arrow points to a label on the bottom line which reads, “C s F (s).”

There are a few issues with this:

- Sentence 3 introduces the lowest line, but without giving it a label. The label is only mentioned in the final sentence. It would help orient the reader if the label was given on first mention.
- Sentences 4 and 5 describe the “Overall charge” label as being on an arrow. It is actually on a curly brace, or perhaps you just call it a bracket.
- Sentences 6, 7, and 8 discuss delta H labels with subscripts 1, 2, and 3. There are no such labels on the current diagram.
- Sentences 6, 7, and 8 discuss the three horizontal lines on the left side with their labels, and then go back and discuss the three arrows with their labels. Wouldn’t it be easier to understand if each line and arrow were discussed in order?
- The arrow colors are not given. The lower right arrow is in a different color, so I suppose they might matter, and it might not hurt to include them.
- There is some explanatory label text, starting with “(Enthalpy of formation),” which is not on the diagram. They are nice clarifiers, but I do not know what the best practice is on including extra information in alt text. I have kept these phrases in my suggested replacement below, but used **gray text** so they are easy to find and remove if you want to.

One possibility is to replace the whole alt text with something like this:

A diagram is shown. An upward facing thin black arrow is drawn to the far left of the chart and is labeled “H increasing.” A horizontal line is drawn at the bottom of the chart labeled, “C s F (s).” To the left side of this line is a bracket extending up to another horizontal line. The bracket is labeled, “Overall change.” Beside this bracket is another label, “capital delta H subscript f, degree sign, equals negative 553.5 k J per mol, (**Enthalpy of formation**).” The horizontal line at the top of the bracket is labeled, “C s (s), plus sign, one half F subscript 2, (g).” From that line, an upward facing red arrow is labeled, “capital delta H subscript s, degree sign, equals 76.5 k J per mol, (**Enthalpy of sublimation**).” The arrow ends at a horizontal line labeled “C s, (g), plus sign, one half F subscript 2, (g).” From that line, an upward facing red arrow is labeled, “I E, equals 375.7 k J per mol, (**ionization energy**).” The arrow ends at a horizontal line labeled, “C

s, superscript positive sign, (g), plus sign, one half F subscript 2, (g).” From that line, an upward facing red arrow is labeled, “one half D equals 79.4 k J / mol, (one half dissociation energy).” The arrow ends at a horizontal line across the top of the diagram labeled “C s, superscript positive sign, (g), plus sign, F, (g).” From the right side of that line, a downward facing red arrow is labeled, “E A equals negative 328.2 k J / mol, (electron affinity).” That arrow ends on a horizontal line labeled, “C s superscript positive sign, (g) plus F superscript negative sign, (g).” From that line, a downward facing purple arrow is labeled, “negative capital delta H subscript lattice equals negative 756.9 k J / mol.” That arrow points to the bottom line, the one labeled, “C s F (s).”

However, that still discusses the lines and arrows in a different order than the text paragraph below the figure discusses them. Reordering the alt text to describe the diagram in the same order that the main text covers it gives us something like this:

A diagram is shown. An upward facing thin black arrow is drawn to the far left of the chart and is labeled “H increasing.” A horizontal line is drawn across the left half of the diagram not quite half way from the bottom to the top. That line is labeled, “C s (s), plus sign, one half F subscript 2, (g).” From that line, an upward facing red arrow is labeled, “capital delta H subscript s, degree sign, equals 76.5 k J per mol, (Enthalpy of sublimation).” The arrow ends at a horizontal line labeled “C s, (g), plus sign, one half F subscript 2, (g).” From that line, an upward facing red arrow is labeled, “I E, equals 375.7 k J per mol, (ionization energy).” The arrow ends at a horizontal line labeled, “C s, superscript positive sign, (g), plus sign, one half F subscript 2, (g).” From that line, an upward facing red arrow is labeled, “one half D equals 79.4 k J / mol, (one half dissociation energy).” The arrow ends at a horizontal line across the top of the diagram labeled “C s, superscript positive sign, (g), plus sign, F, (g).” From the right side of that line, a downward facing red arrow is labeled, “E A equals negative 328.2 k J / mol, (electron affinity).” That arrow ends on a horizontal line labeled, “C s superscript positive sign, (g) plus F superscript negative sign, (g).” From that line, a downward facing purple arrow is labeled, “negative capital delta H subscript lattice equals negative 756.9 k J / mol.” That arrow ends at the right side of a horizontal line across the bottom of the diagram labeled, “C s F (s).” To the left side of this line is a bracket extending up to the horizontal line that was described first. The bracket is labeled, “Overall change.” Beside this bracket is another label, “capital delta H subscript f, degree sign, equals negative 553.5 k J per mol, (Enthalpy of formation).” Again, the horizontal line at the top of the bracket is labeled, “C s (s), plus sign, one half F subscript 2, (g).”

Section The Born-Haber Cycle, 1st paragraph after Figure 7.13

This paragraph describes the Born-Haber cycle, seemingly in reference to figure 7.13. This is helpful, but as a reader it was harder to get oriented since the first step described is neither at the top nor the bottom of the figure. Would it help to add something like this to the start of sentence 2?

Starting at the bottom, left red arrow in Figure 7.13, ... (the remainder of sentence 2)
There may be better ways to word it.

Table 7.4 Formatting on Firefox

The formatting of table 7.4 is off on Firefox, at least on Windows 11.

First, for comparison, here is the top right corner of the table 7.4 on Chrome 146.0.7680.178 (64 bit) on Windows 11, where it is **working correctly**.

$\Delta H = \Delta H_s^\circ$ $= 76.5 \text{ kJ/mol}$
$\Delta H = \frac{1}{2} D$ $= 79.4 \text{ kJ/mol}$
$\Delta H = IE$ $= 375.7 \text{ kJ/mol}$

Now, here is the top right corner of table 7.4 on Firefox 149.0 (64-bit) on Windows 11, where it is **not working**. The last column text extends outside the table to the right.

$\Delta H = \Delta H_s^\circ = 76.5 \text{ kJ/mol}$
$\Delta H = \frac{1}{2} D = 79.4 \text{ kJ/mol}$
$\Delta H = IE = 375.7 \text{ kJ/mol}$

This issue also results in a horizontal scroll bar displaying at the bottom of the table 7.4 on Firefox. Here are screenshots of the bottom of table 7.4 on Firefox scrolled left and right.

Enthalpy of formation of CsF(s), add steps 1–5	$\Delta H = \Delta H_f^\circ = \Delta H_s^\circ + \frac{1}{2} D + IE + (EA) + (-\Delta H_{\text{lattice}})$ $\text{Cs}(s) + \frac{1}{2} \text{F}_2(g) \rightarrow \text{CsF}(s)$	$\Delta H = -553.5 \text{ kJ/mol}$
--	--	------------------------------------

Table 7.4

Enthalpy of formation of CsF(s), add steps 1–5	$\Delta H = \Delta H_f^\circ = \Delta H_s^\circ + \frac{1}{2} D + IE + (EA) + (-\Delta H_{\text{lattice}})$ $\text{Cs}(s) + \frac{1}{2} \text{F}_2(g) \rightarrow \text{CsF}(s)$	$\Delta H = -553.5 \text{ kJ/mol}$
--	--	------------------------------------

Chapter 7.6

Figure 7.16 alt text

A more accessible approach might be to put all of this in a real HTML table, with separate images in each individual cell where the cell contents are graphical. That way each alt text description would be shorter and more contained. Maybe that is difficult to implement, though. This same suggestion would also apply to figure 7.19.

The alt text currently reads:

A table with four rows and six columns is shown. The header column contains the phrases, “Number of regions,” “Spatial arrangement,” “Wedge/dash Notation,” and “Electron region Geometry.” The first row reads: “Two regions of high electron density (bonds and/or unshared pairs),” “Three regions of high electron density (bonds and/or unshared pairs),” “Four regions of high electron density (bonds and/or unshared pairs),” “Five regions of high electron density (bonds and/or unshared pairs),” and “Six regions of high electron density (bonds and/or unshared pairs).” The second row shows diagrams of orbitals. The first image shows two oval-shaped orbs with an arrow indicating an angle of 180 degrees. The second image shows three oval-shaped orbs with an arrow indicating an angle of 120 degrees. The third image shows four oval-shaped orbs with an arrow indicating an angle of 109.5 degrees. The fourth image shows five oval-shaped orbs with an arrow indicating an angle of 90 and 120 degrees. The fifth image shows six oval-shaped orbs with an arrow indicating an angle of 90 degrees. The third row contains Lewis structures. The first structure shows a beryllium atom single bonded to two hydrogen atoms. The second structure shows a boron atom single bonded to three hydrogen atoms. The third structure shows a carbon atom single bonded to four hydrogen atoms. The fourth structure shows a phosphorus atom single

bonded to five fluorine atoms. The fifth structure shows a sulfur atom single bonded to six fluorine atoms. The fourth row contains the phrases “Linear; 180 degree angle,” “Trigonal Planar; all angles 120 degrees,” “Tetrahedral; all angles 109.5 degrees,” “Trigonal bipyramidal; angles of 90 degrees and 120 degrees. An attached atom may be equatorial, (in the plane of the triangle), or axial, (above the plane of the triangle),” and “Octahedral; 90 degrees or 180 degrees.”

Issues:

- Sentence 2 says the third row header column reads “**Wedge/dash Notation**” when it actually reads “**Line-dash-wedge notation**”.
- The descriptions of spatial arrangements in row two never explicitly say where the ovals meet. This might make it a little harder to conceptualize what an angle between ovals means. Perhaps mentioning that they meet in the center might help, even if it was only mentioned for the first row 2 image in sentence 5.
- Sentence 8 describes “an arrow” with “an angle” with two different degree measurements, 90 and 120. Furthermore, there is a bit more structure to the arrangement of the orbitals in this image compared to the preceding images. I think that can be reasonably expressed.
- Sentence 9 could describe the image with six orbitals more explicitly. Also, pedantically, there are two arrows not one.
- Sentences 10 through 15 describe the third row. Although the row’s header column is “Line-dash-wedge” notation, and the whole table is about geometries, neither the wedges nor the dashes are described.
- Towards the end, the opening quotation mark is missing before the phrase **Trigonal Planar; all angles 120 degrees**, and the phrase “90 degrees **and** 120 degrees” should be “90 degrees **or** 120 degrees”.
- Toward the end, there is a single quoted phrase in a list of quoted phrases that is very long, contains its own list, and even contains its own period (which is why I didn’t give sentence numbers for these last bullet points, does that count?) Anyway, the phrase is so long I forgot where I was by the end, and it was hard to rejoin the original outer list of quoted phrases. I wonder if that could all be broken up better in the alt text description:
“Trigonal bipyramidal; angles of 90 degrees and 120 degrees. An attached atom may be equatorial, (in the plane of the triangle), or axial, (above the plane of the triangle),”
- At the end, the phrase “Octahedral; 90 degrees or 180 degrees.” should be “Octahedral; **all angles** 90 degrees or 180 degrees.”
- Overall, in a long linear description described by row, it is hard to remember which column you are in, especially in row four. The other rows have numbers of ovals or bonds that match the number of regions of high electron density for that column. However, row four has nothing like that, and the column descriptions are long past. Perhaps noting which column each cell of row four is in would be helpful.
- Finally, why are the row headings in the left column only listed in the beginning of the alt text, requiring the reader to remember them through the whole description? I suggest giving each row’s heading at the beginning of that row’s description.

Here is a suggested edit of the alt text of figure 7.16 trying to address these issues:

A table with four rows and six columns is shown. The header column contains the phrases, "Number of regions," "Spatial arrangement," "**Line-dash-wedge notation**," and "Electron region **geometry**." The "**Number of regions**" row reads: "Two regions of high electron density (bonds and/or unshared pairs)", "Three regions of high electron density (bonds and/or unshared pairs)", "Four regions of high electron density (bonds and/or unshared pairs)", "Five regions of high electron density (bonds and/or unshared pairs)", and "Six regions of high electron density (bonds and/or unshared pairs)." The "**Spatial arrangement**" row shows diagrams of orbitals. The first image shows two oval-shaped orbs **that meet at a center point with a curving** arrow indicating an angle of 180 degrees. The second image shows three oval-shaped orbs with an arrow indicating an angle of 120 degrees. The third image shows four oval-shaped orbs with an arrow indicating an angle of 109.5 degrees. **The fourth image shows five oval-shaped orbs. Three of the orbs appear to be in a plane. The remaining two orbs stick up and down perpendicular to that plane. An arrow indicates that the angle between two of the three orbs in the plane is 120 degrees. A second arrow indicates that the angle between one of the orbs in the plane and one of the orbs sticking out of the plane is 90 degrees. The fifth image shows six oval-shaped orbs arranged along both directions on all three axes. Arrows show that the angles between the orbs are 90 degrees.** The "**Line-dash-wedge notation**" row contains Lewis structures. The first structure shows a beryllium atom single bonded to two hydrogen atoms. The second structure shows a boron atom single bonded to three hydrogen atoms. The third structure shows a carbon atom single bonded to four hydrogen atoms. **Two of the bonds are shown as single lines. One of the bonds is shown as a solid wedge coming in front of the plane of the paper. One of the bonds is shown as a dashed wedge going behind the plane of the paper.** The fourth structure shows a phosphorus atom single bonded to five fluorine atoms. **Three of the bonds are lines, one bond is a solid wedge, and one bond is a dashed wedge.** The fifth structure shows a sulfur atom single bonded to six fluorine atoms. **The top and bottom bonds are lines, The lower left and lower right bonds are solid wedges. The upper left and upper right bonds are dashed wedges.** The "**Electron region geometry**" row contains the following phrases: "Linear; 180 degree angle," **for the two region column**, "**Trigonal Planar; all angles 120 degrees,**" **for the three region column**, "Tetrahedral; all angles 109.5 degrees," **for the four region column. The five region column has the phrase** "Trigonal bipyramidal; angles of 90 degrees or 120 degrees. An attached atom may be equatorial, (in the plane of the triangle), or axial, (above the plane of the triangle)." **The six region column has the phrase** "Octahedral; all angles 90 degrees or 180 degrees."

Figure 7.17 caption

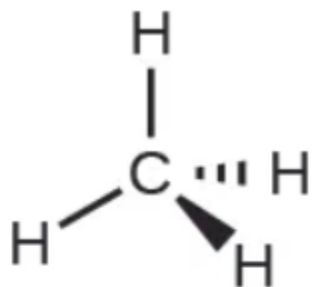


Figure 7.17 The molecular structure of the methane molecule, CH_4 , is shown with a tetrahedral arrangement of the hydrogen atoms. VSEPR structures like this one are often drawn using the wedge and dash notation, in which solid lines represent bonds in the plane of the page, solid wedges represent bonds coming up out of the plane, and dashed lines represent bonds going down into the plane.

There are a couple of issues with this caption:

- The caption describes the wedge and dash notation. It is a nice, clear description. However, if the wedge and dash is a notation that will be reused later in the book, shouldn't its description be in the main text instead of being in a caption?
- The narrow formatting on the long caption looks a bit odd here. I understand that figure captions are consistently the width of their figures, but I thought I'd mention it. Moving the wedge and dash notation out of the caption would have the secondary advantage of shortening this caption so the formatting looks less out of place.

Figure 7.18 alt text

The alt text reads:

Three images are shown and labeled, “a,” “b,” and “c.” Image a shows a nitrogen atom single bonded to three hydrogen atoms. There are four oval-shaped orbs that surround each hydrogen and one facing away from the rest of the molecule. These orbs are located in a tetrahedral arrangement. Image b shows a ball-and-stick model of the nitrogen single bonded to the three hydrogen atoms. Image c is the same as image a, but there are four curved, double headed arrows that circle the molecule and are labeled, “106.8 degrees.”

Some issues:

- The image a description says there are three hydrogen atoms, which are not shown on the image. Then it says that there are four orbs surrounding each hydrogen atom. If you interpret that as four orbs per each of the three hydrogen atoms it sounds like there are 12 orbs. If you interpret it as one orb per hydrogen atom it doesn't make sense because there are 4 orbs but only 3 hydrogen atoms.
- The image a description does not mention the “lone pair” label or the dots.
- The image b description misses the main point of the figure. If I understand correctly, the point is that the influence of the lone pair occupying the top orbital in the tetrahedron results in a ball-and-stick model that does not look like three bonds in a plane, but like the bottom of a tetrahedron with the top apparently missing. I'm sure there is a better way to phrase it, but the existing description does not address the spatial layout of the bonds at all.
- The image c description makes it sound like all the arrows are 106.8 degrees. I think the label only refers to the one arrow, because the caption says that the top “lone pair takes up a larger region of space”

Here is a suggested reworking of the figure 7.17 alt text:

Three images are shown and labeled, “a,” “b,” and “c.” Image a shows a nitrogen atom surrounded by four oval-shaped orbitals in a tetrahedral arrangement around a central dot. The top orbital has two black dots at the end of it labeled “Lone pair.” Image b shows a ball-and-stick model of a blue nitrogen atom single bonded to three white hydrogen atoms. The nitrogen atom is in the position of the central dot from image a, and the three hydrogen bonds are in the locations of the three orbitals from image a that did not have the lone pair of dots. This results in the ball-and-stick model being shaped like a triangular pyramid with the blue nitrogen atom at the top. The region that held the upward pointing orbital with the lone pair in image a is empty in image b. Image c is the same as image a, but there are four curved, double headed arrows that circle the molecule. An arrow between two of the bottom orbitals is labeled, “106.8 degrees.”

Figure 7.19 alt text

Like figure 7.16, this figure might be more accessible if it was replaced with an HTML table and used images within the cells of that table.

To make things more explicit, I suggest replacing “on each side” with “on both sides” everywhere it appears, specifically in sentences 4, 19, 31, 40, and 46.

Sentences 13 and 14 read:

The fourth space in the second column contains a structure in which the letter E is single bonded to the letter X on five sides. The angle between the bonds is labeled with a curved, double headed arrow and the values “90 and 120 degrees.”

Technically, “with a curved, double headed arrow” should read “with curved, double headed arrows”, plural.

Sentences 19 and 20 read (emphases mine):

The first space in the third column is empty while the second contains a structure in which the letter E is single bonded to the letter X on each side **and has** a lone pair of electrons. **The angle between the bonds** is labeled with a curved, double headed arrow and the value, “less than 120 degrees.”

There is nothing wrong with these sentences, but I include them for reference when discussing sentences 22 and 23 below.

Sentences 22 and 23 read (emphases mine):

The third space in the third column contains a structure in which the letter E is single bonded to the letter X three times **and to** a lone pair of electrons. **It** is labeled with a curved, double headed arrow and the value, “less than 109 degrees.”

The end of sentence 22 implies that the letter E not only has a lone pair of electrons, but is bonded to a lone pair of electrons. I am not sure a “bond” is the correct terminology for that relationship. More importantly, it leaves sentence 23 ambiguous when discussing the curved arrow labeled “less than 109 degrees.” Does “less than 109 degrees” refer to the angle between the bonds with other atoms? Or does it refer to the angle between a bond with an atom and the “bond” with the lone pair of electrons? Or both. Hopefully the reader would garner the fact that it was the angle between bonds with other atoms, but the text could be less ambiguous and more explicit. I suggest making sentence 22 and 23 more like sentences 19 and 20 (changes in bold):

The third space in the third column contains a structure in which the letter E is single bonded to the letter X three times **and has** a lone pair of electrons. **The angle between the bonds** is labeled with a curved, double headed arrow and the value, “less than 109 degrees.”

Sentence 34 below has a similar issue.

Sentences 25 and 26 read (emphases mine):

The fourth space in the third column contains a structure in which the letter E is single bonded to the letter X on four sides and has a lone pair of electrons. The bond **angle is** labeled with **a** curved, double headed **arrow** and the values, “less than 90 **and** less than 120 degrees.”

This suggests there is only one arrow when there are two, and that there is only one label when there are two. Suggest improving this to (changes in bold):

The fourth space in the third column contains a structure in which the letter E is single bonded to the letter X on four sides and has a lone pair of electrons. The bond **angles**

are labeled with curved, double headed **arrows** and the values, “less than 90 **degrees**,” **and** “less than 120 degrees.”

Sentence 30 reads:

The structure is labeled, “Square pyramidal.”

In fact, it is labeled just “Square pyramid” with no “-al” suffix. Suggest:

The structure is labeled, “**Square pyramid**.”

Sentence 32 reads:

The bond angle is labeled with a curved, double headed arrow and the value, “less than less than 109 degrees.”

Doesn't the math symbol $< <$ mean “**much less than**”? Would that be clearer than “less than less than”? Right now it reads like a typo, even though it is not.

Sentences 34 and 35 read (emphasis mine):

The fourth space in the fourth column contains a structure in which the letter E is single bonded to the letter X three times **and to** two lone pairs of electrons. The bond angle is labeled with a curved, double headed arrow and the value, “less than 90 degrees.”

This has the same issue as sentence 22 above. It implies that the E is “bonded to” the lone pairs of electrons, leaving the next sentence ambiguous as to what angle is being referred to. I suggest reusing the phrasing sentence 19 and most other sentences. Sentence 35 is already good in this case. I only included it for context. I suggest replacing with:

The fourth space in the fourth column contains a structure in which the letter E is single bonded to the letter X three times **and has** two lone pairs of electrons. The bond angle is labeled with a curved, double headed arrow and the value, “less than 90 degrees.”

Sentences 37 and 38 read (emphasis mine):

The fifth space in the fourth column contains a structure in which the letter E is single bonded to the letter X on four sides and has two lone pairs of electrons. The **bond angle** is labeled with a curved, double headed arrow and the value “90 degrees.”

If I understand correctly, it is required here for the two lone pairs of electrons are on opposite sides of the central atom E. If so, I suggest adding that fact (changes in bold):

The fifth space in the fourth column contains a structure in which the letter E is single bonded to the letter X on four sides and has two lone pairs of electrons. **The two lone pairs of electrons are on opposite sides of the letter E.** The **angle between the bonds** is labeled with a curved, double headed arrow and the value “90 degrees.”

Figure 7.20

As someone who is partially red-green colorblind, the red lines are a little hard to see, especially contrasting with the green lines and the green shaded background. Perhaps making them thicker as well as red, and calling them “(thick red lines)” in the caption would help?

Figure 7.20 alt text

The alt text currently reads:

Four sets of images are shown and labeled, “a,” “b,” “c,” and “d.” Each image is separated by a dashed vertical line. Image a shows a six-faced, bi-pyramidal structure where the central vertical axis is labeled, “Axial,” and the horizontal plane is labeled, “Equatorial.” Image b shows a pair of diagrams in the same shape as image a, but in these diagrams, the left has a chlorine atom in the center while the right has a chlorine atom in the center, two fluorine atoms on the upper and lower ends, and one fluorine in the left horizontal position. Image c shows a pair of diagrams in the same shape as image a, but in these diagrams, the left has a chlorine atom in the center while the right has a chlorine atom in the center and three fluorine atoms in each horizontal position. Image d shows a pair of diagrams in the same shape as image a, but in these diagrams, the left has a chlorine atom in the center while the right has a chlorine atom in the center, two fluorine atoms in the horizontal positions, and one in the axial bottom position.

The current alt text has the following features:

- It does not describe the shape very fully for part (a), fixed below.
- It does not give the locations of the red lines mentioned in the caption, fixed below. Figures 7.23, 7.24, and 7.25 have the same issue.
- It does not mention the green lines from the chlorine to the fluorine atoms that connect them like a molecular structure diagram. I do not know if discussing these would actually make it clearer, so I did **not** address this one below.

I suggest changing the alt text to the following (changes in bold):

Four sets of images are shown and labeled, “a,” “b,” “c,” and “d.” Each image is separated by a dashed vertical line. Image a shows a six-faced, **five-pointed** bi-pyramidal structure where the central vertical axis is labeled, “Axial,” and the horizontal plane is labeled, “Equatorial.” **The horizontal plane has three points and forms the triangular base of both the top and bottom pyramids. The vertical axis ends in two points which are the points of two pyramids above and below the horizontal plane.** Image b shows a pair of diagrams in the same shape as image a, but in these diagrams, the left has a chlorine atom in the center **and two red lines from the center to the two right horizontal positions**, while the right has a chlorine atom in the center, two fluorine atoms on the upper and lower ends, and one fluorine in the left horizontal position. Image c shows a pair of diagrams in the same shape as image a, but in these diagrams, the left has a chlorine atom in the center **and two red lines from the center to the upper and lower axial position**, while the right has a chlorine atom in the center and three fluorine atoms in each horizontal position. Image d shows a pair of diagrams in the same shape as image a, but in these diagrams, the left has a chlorine atom in the center **and two red lines from the center to the axial top position and one of the right horizontal positions**, while the right has a chlorine atom in the center, two fluorine atoms in the **other two** horizontal positions, and one in the axial bottom position.

Figure 7.22 alt text

The last sentence of the alt text reads (emphasis mine):

This figure uses dashes and wedges to **displays its three planes** in a tetrahedral shape.

I believe “displays” should be the singular “**display**” instead. Also, what does “its three planes” mean? Should it be “the three dimensions”? A tetrahedron has four surface planes (counting the bottom). If you count planes passing through the tetrahedron’s center point, I think there would be more than three because you could do it using any of the outer points as the “apex”.

Figure 7.23 alt text

The alt text reads:

Two diagrams are shown and labeled, “a” and “b.” Diagram a shows an oxygen atom in the center of a four-sided pyramid shape. Diagram b shows the same image as diagram a, but this time there are hydrogen atoms located at two corners of the pyramid shape.

This does not mention the red lines on the left diagram. Figure 7.20 had the same issue. Here is possible replacement alt text (changes in bold):

Two diagrams are shown and labeled, “a” and “b.” Diagram a shows an oxygen atom in the center of a four-sided **triangular** pyramid shape. **Two red lines connect the center oxygen to the two corners to the lower right.** Diagram b shows the same image as diagram a, but this time there are hydrogen atoms located at **upper and left** two corners of the pyramid shape.

Figure 7.24 alt text

The alt text reads:

Two diagrams are shown and labeled, “a” and “b.” Diagram a shows a sulfur atom in the center of a six-sided bi-pyramidal shape. Diagram b shows the same image as diagram a, but this time there are fluorine atoms located at four corners of the pyramid shape and they are connected to the sulfur atom by single lines.

Like the figures 7.20 and 7.23 alt text, this does not describe the red line. Nor does it describe that the points with the fluorine atoms avoid the point with the red line. Suggested replacement (changes in bold):

Two diagrams are shown and labeled, “a” and “b.” Diagram a shows a sulfur atom in the center of a six-sided, **five-pointed** bi-pyramidal shape **with a red line connecting the sulfur to the leftmost of the three equatorial corners of the shape.** Diagram b shows the same image as diagram a, but this time there are fluorine atoms located at **the other** four corners of the pyramid shape and they are connected to the sulfur atom by single lines.

Figure 7.25 alt text

The alt text reads:

Two diagrams are shown and labeled, “a” and “b.” Diagram a shows a xenon atom in the center of an eight-sided octahedral shape. Diagram b shows the same image as diagram a, but this time there are fluorine atoms located at the four corners of the shape in the horizontal plane. They are connected to the xenon by single lines.

Like the alt text for figures 7.20, 7.23, and 7.24, this does not describe the red lines. Here is a suggested replacement (changes in bold):

Two diagrams are shown and labeled, “a” and “b.” Diagram a shows a xenon atom in the center of an eight-sided, **six-pointed** octahedral shape. **Red lines connect the xenon to the top and bottom points of the shape.** Diagram b shows the same image as diagram a, but this time there are fluorine atoms located at the four corners of the shape in the horizontal plane. They are connected to the xenon by single lines.

Unlabeled image in Example 7.16, Solution section, alt text

Sentence 2 reads (emphasis mine):

The atoms described are drawn with bonds that indicate a three-dimensional, **tetrahedral** shape around the nitrogen atom.

Since this shape description sentence just talks about the bonds not the lone pair of electrons, shouldn't the shape described match the molecular shape of trigonal pyramidal (from the solution answer) instead of the electron pair geometry of tetrahedral (again from the solution answer.) Here is a suggested replacement (change in bold):

The atoms described are drawn with bonds that indicate a three-dimensional, **trigonal pyramid** shape around the nitrogen atom.

Note: Sentence 3 of the same alt text does **not** have this issue, because the “tetrahedral” in sentence 3 matches the molecular shape given for that local portion in the solution.

Unlabeled image in Example 7.16, Check Your Learning section, alt text

The alt text reads:

A Lewis structure depicts a nitrogen atom with two lone pairs of electrons that is single bonded to two hydrogen atoms and a carbon atom, which is, in turn, single bonded to a hydrogen atom, a methyl group and another carbon atom. This carbon atom is single bonded to an oxygen atom with two lone pairs of electrons and single bonded to an oxygen that has two lone pairs of electrons and a single bond to a hydrogen atom.

There are a couple of issues:

- The nitrogen atom only has one lone pair of electrons, not two.
- At this point in the book, students haven't been introduced to methyl groups.
- The right-most carbon atom is not single bonded to two oxygen atoms, it is double bonded to one and single bonded to the other.

Here is a suggested replacement (changes in bold):

A Lewis structure depicts a nitrogen atom with **one** lone **pair** of electrons that is single bonded to two hydrogen atoms and a carbon atom, which is, in turn, single bonded to a hydrogen atom, a **C H subscript 3** methyl group, and another carbon atom. This carbon atom is **double** bonded to an oxygen atom with two lone pairs of electrons and single bonded to **another** oxygen that has two lone pairs of electrons and a single bond to a hydrogen atom.

Example 7.17, section Molecular Simulation, paragraph 1

The end of the last sentence of paragraph 1 reads (emphasis mine):

... and/or molecular structure (called “molecular **shape**” in the simulator).

The label in the simulator is “molecular **geometry**”, not “molecular shape”. (It may be “molecular shape” if you use an old Java version?)

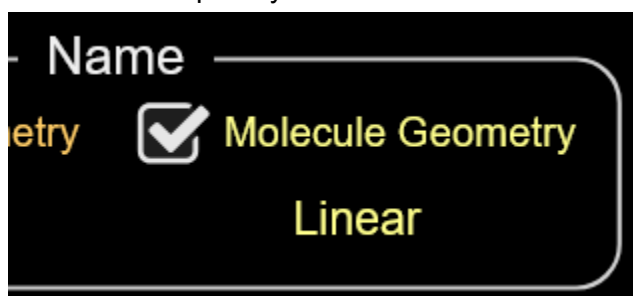


Figure 7.26 alt text

The alt text reads:

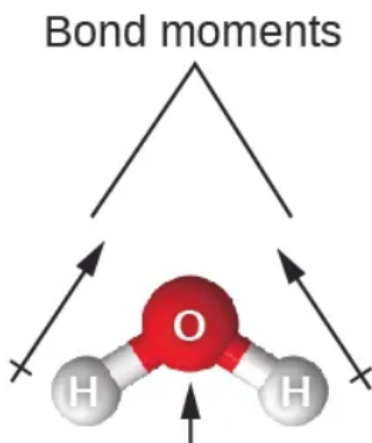
Two images are shown and labeled, “a” and “b.” Image a shows a large sphere labeled, “C,” a left-facing arrow with a crossed end, and a smaller sphere labeled “H.” Image b shows a large sphere labeled, “B,” a right-facing arrow with a crossed end, and a smaller sphere labeled “F.”

I think this could be a little clearer, especially since this is the first description of the shape of the dipole vector. I also wonder about depending on the order in which atoms are mentioned to convey left/right information if the text is ever translated for use by right-to-left or vertical language students. Here is a suggested replacement (changes in bold):

Two images are shown and labeled, “a” and “b.” Image a shows a large sphere labeled, “C” **on the left** and a smaller sphere labeled “H” **to the right. Above it is** a left-facing arrow with a crossed end **on the right end**. Image b shows a large sphere labeled, “B” **to the left** and a smaller sphere labeled, “F” **to the right. Above it is a longer** right-facing arrow with a crossed end **on the left end**.

Figure 7.27

In part (b) of the figure, shouldn't the two top arrows labeled "Bond moments" be parallel with the bonds between the hydrogen atoms and the oxygen atom? In other words, shouldn't this:



Look more like this:

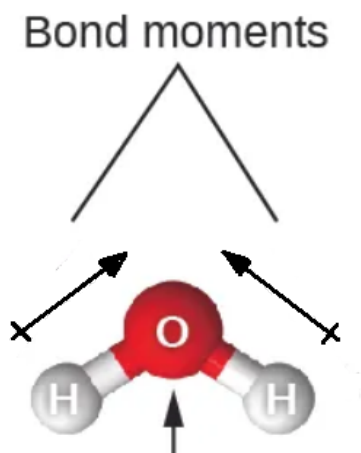


Figure 7.27 alt text

The alt text reads:

Two images are shown and labeled, "a" and "b." Image a shows a carbon atom bonded to two oxygen atoms in a ball-and-stick representation. Two arrows face away from the center of the molecule in opposite directions and are drawn horizontally like the molecule. These arrows are labeled, "Bond moments," and the image is labeled, "Overall dipole moment equals 0." Image b shows an oxygen atom bonded to two hydrogen atoms in a downward-facing v-shaped arrangement. An upward-facing, vertical arrow is drawn below the molecule while two upward and inward facing arrows are drawn above the molecule. The upper arrows are labeled, "Bond moments," while the image is labeled, "Overall dipole moment."

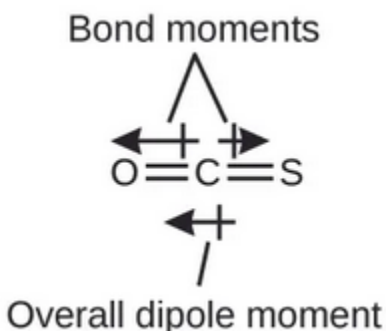
Issues:

- It doesn't mention the linear arrangement of molecule a, but does mention the v-shaped arrangement of molecule b.
- It doesn't mention that the arrow lengths in molecule a are equal to each other, something which distinguishes Figure 7.27 (a) from unlabeled OCS molecule image below it.
- Describing the overall molecule dipole moment arrow last would keep that description next to the image label "Overall dipole moment."
- I believe the "Overall dipole moment" describes the bottom arrow in (b), not the entire figure part (b)

Here is a suggested replacement (changes in bold):

Two images are shown and labeled, "a" and "b." Image a shows a carbon atom bonded to two oxygen atoms in a ball-and-stick representation. **The molecule is in a straight line linear shape.** Two arrows **of equal length** face away from the center of the molecule in opposite directions. **They** are drawn horizontally like the molecule. These arrows are labeled, "Bond moments," and the image is labeled, "Overall dipole moment equals 0." Image b shows an oxygen atom bonded to two hydrogen atoms in a downward-facing v-shaped arrangement **with the oxygen in the center toward the top.** Two upward and inward facing arrows are drawn above the molecule **in roughly the directions from the hydrogen atoms to the oxygen atom.** These arrows are labeled, "Bond moments." **Below the molecule is an upward-facing, vertical arrow.** **This arrow** is labeled, "Overall dipole moment."

1st unlabeled figure below labeled figure 7.27, alt text

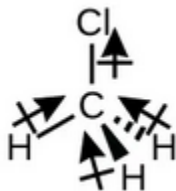


The first sentence of the alt text reads (emphasis mine):

An image shows a carbon atom double bonded to a sulfur atom and an oxygen atom which are arranged in a horizontal **plane**.

In place of the word "plane", wouldn't the word "**line**" be a better description. There is nothing 2-D about the arrangement. Also, if it was in a plane, that would allow a bend, and the fact that there is an overall dipole moment due to the differing bond moment magnitudes despite there being no bend seems to be the point of the figure.

2st unlabeled figure below labeled figure 7.27, alt text

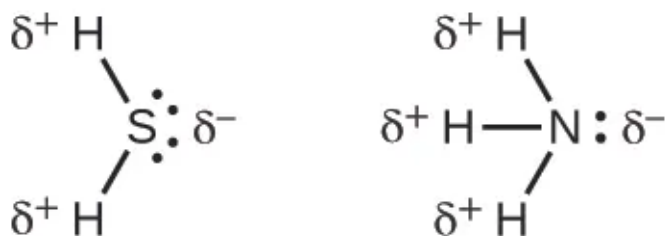


Sentence 3 reads:

The carbon and chlorine arrow is longer.

I don't think it is longer. In fact, the plus sign cross mark appears closer to the arrowhead, meaning it is actually slightly shorter. I suggest correcting or removing sentence 3.

3rd unlabeled figure below labeled figure 7.27



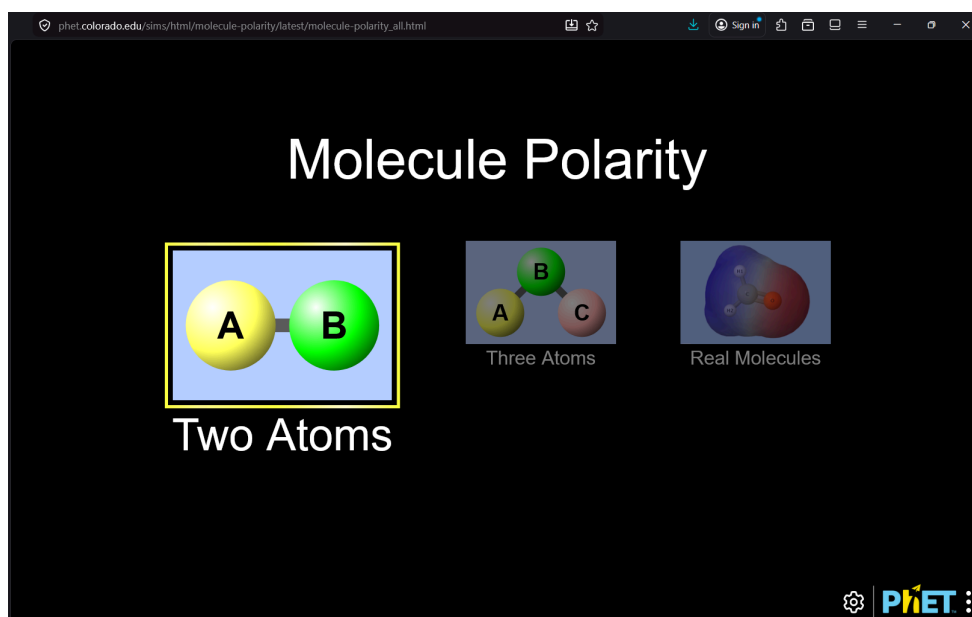
I do not believe the lower case delta notation has been officially introduced, at least not at this point in the book. I assume these are partial charges, but it might be good to officially describe the nomenclature.

Example 7.18, section Polarity Simulations, paragraph 1

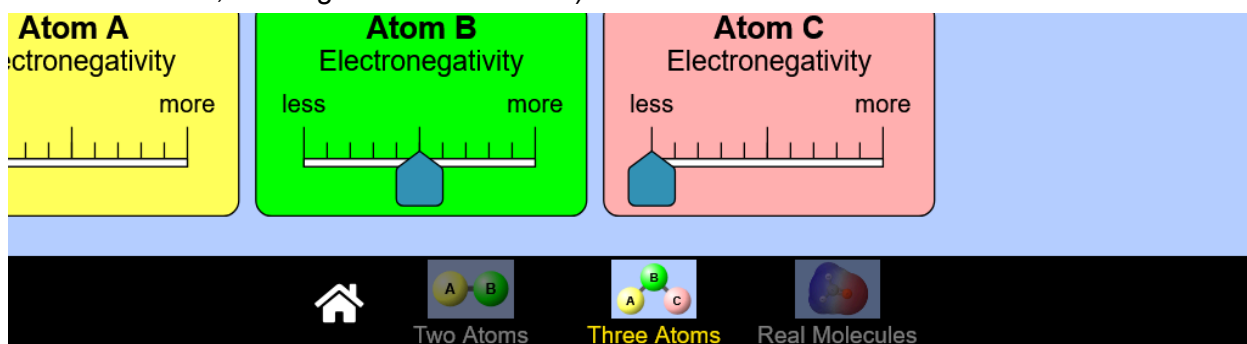
Paragraph 1, sentence 1 reads (emphasis mine):

Open the molecule polarity simulation and select the "Three Atoms" **tab at the top**.

At least in my Firefox browser on Windows, the "Three Atoms" looks more like a button or an image than a tab. Also, on the Molecular Polarity home screen, the buttons are across the middle, not the top. (If you use the old Java version, it might look like a tab.)



After going into one of the options, the buttons are along the bottom, not the top. (If you use the old Java version, this might not be the case.)



Since the buttons change position, perhaps it is best not to mention the position at all, especially since they are so prominent on the initial home screen.

Therefore, I suggest replacing paragraph 1, sentence 1 with (change in bold):

Open the molecule polarity simulation and select the “Three Atoms” **button**.

Chapter 7 - Key Terms

Lewis structure

I noticed that the definition of key term **electron-pair geometry** included not only bonds and lone pairs, but also unpaired electrons. Should the **Lewis structure** definition also include unpaired electrons? Compare with the Chapter 7 Summary, section 7.3, sentence 2 description of a Lewis structure, which includes unpaired electrons:

Lone pairs, unpaired electrons, and single, double, or triple bonds are used to indicate where the valence electrons are located around each atom in a Lewis structure.

molecular structure

It seems odd that it appears as a key term twice. Should they be combined? If not, should the second **molecular structure** definition end with “(compare with **electron-pair geometry**)” and the **electron-pair geometry** definition end with “(compare with **molecular structure**)”? That is what the second molecular structure definition is getting at with the word “only”, correct? Just a thought.

Chapter 7 - Exercises

Note: only odd numbered exercises are fully reviewed here, as I do not have instructor access. I was able to do some alt text review of even numbered questions, however.

Exercise 7.7 solution

This may make no difference, but the orbital orderings in the answers are inconsistent. For example, in the answer for (a), $[\text{Ar}]4s^23d^{10}4p^6$, 4s comes before 3d, whereas in the answer for (b), $[\text{Kr}]4d^{10}5s^25p^6$, 5s comes after 4d. Answers to the other parts may be minorly inconsistent in this way as well.

Exercise 7.25 solution

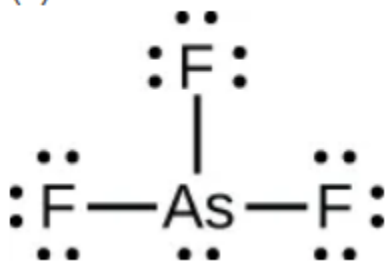
The 7.25 question starts (emphases mine):

Write the **Lewis symbols of the ions** in each of the following ionic compounds and the **Lewis symbols of the atom from which they are formed**: ...

However, the 7.25 solution only includes Lewis symbols for the ions. It does not include Lewis symbols for the atoms from which those ions were formed.

Exercise 7.29 (c) answer, unlabeled figure alt text

(c)



The alt text reads (emphasis mine):

A Lewis structure shows an arsenic atom single bonded to three fluorine atoms. Each fluorine atom **has a lone pair** of electrons.

The final sentence should say “**has three lone pairs**”, not “has a lone pair”.

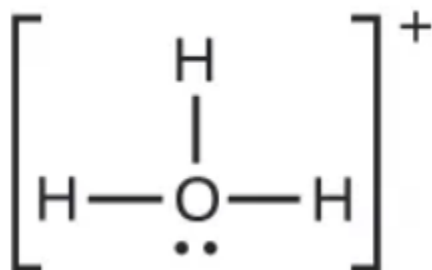
Exercise 7.29 (k) answer, unlabeled figure

(k)



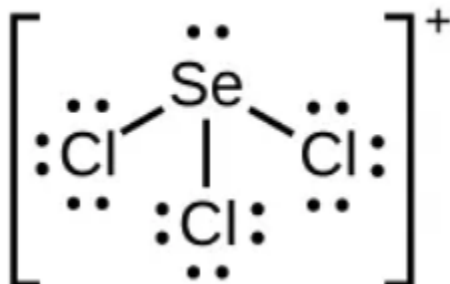
Should this Lewis structure image have brackets around the whole structure and the superscripted 2+ outside of the brackets? This would make it clear that the 2+ applies to the entire structure. It would also be more consistent with the answer images for other parts of exercise 7.29, such as 7.29(f) below. **Note:** If the image changes, the alt text should as well, as the brackets in the other answers are mentioned in their alt texts.

(f)



Exercise 7.31 (c) solution, unlabeled figure alt text

(c) SeCl_3^+ :



The last sentence of the alt text reads:

The whole structure is surrounded by brackets.

This does not mention the superscripted plus sign. I suggest replacing the last sentence (change in bold):

The whole structure is surrounded by brackets **with a superscripted plus sign**.

Exercise 7.33

The 7.33 question reads:

Correct the following statement: "The bonds in solid PbCl_2 are ionic; the bond in a HCl molecule is covalent. Thus, all of the valence electrons in PbCl_2 are located on the Cl^- ions, and all of the valence electrons in a HCl molecule are shared between the H and Cl atoms."

The answer reads (emphases mine):

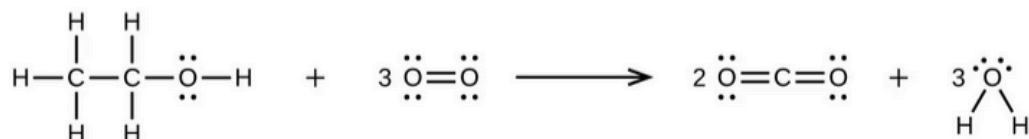
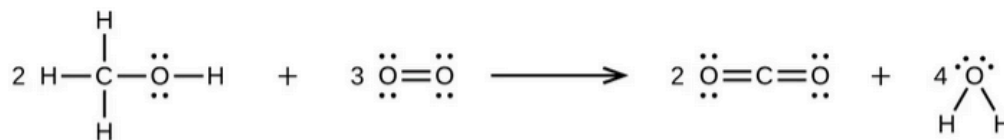
Two valence electrons per Pb atom are transferred to Cl atoms; **the resulting Pb^{2+} ion has a $6s^2$ valence shell configuration**. Two of the valence electrons in the HCl molecule are shared, **and the other six are located on the Cl atom as lone pairs of electrons**.

I found this question confusing in a couple of ways.

- Just from the book, how is a student supposed to know that solid PbCl_2 really is ionic, and that it should not be corrected to covalent. From Table 7.6, Pb has electronegativity 1.9, while Cl has electronegativity 3.0. According to Figure 7.8, a 1.1 electronegativity difference would lead a student to guess that the Pb Cl bond type is polar covalent, not ionic.
- Just from the wording of the question, I had no idea that I was supposed to include the information I bolded in the answer.

Then again, maybe the question is fine and my understanding is lacking.

Exercise 7.35 solution unlabeled image alt text



35.

The first half of the alt text describes the top reaction. However, the figure must have changed because the coefficient numbers in front of the structures in the top reaction have all changed. Here is a corrected first half of the alt text describing the top reaction with the new coefficients (4 changes in bold):

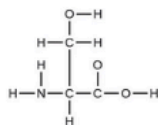
Two reactions are shown using Lewis structures. The top reaction **shows the number two, followed by a carbon atom single bonded** to three hydrogen atoms and single bonded to an oxygen atom with two lone pairs of electrons. The oxygen atom is also bonded to a hydrogen atom. This is followed by a plus sign and the number **three**, followed by two oxygen atoms bonded together with a double bond and each with two lone pairs of electrons. A right-facing arrow leads to **the number two, followed by a carbon atom that is double bonded** to two oxygen atoms, each of which has two lone pairs of electrons. This structure is followed by a plus sign, a number **four**, and a structure made up of an oxygen with two lone pairs of electrons single bonded to two hydrogen atoms.

Exercise 7.39 question formatting

All 5 of the molecular structure diagrams in the exercise 7.39 question are quite small, at least on the web.

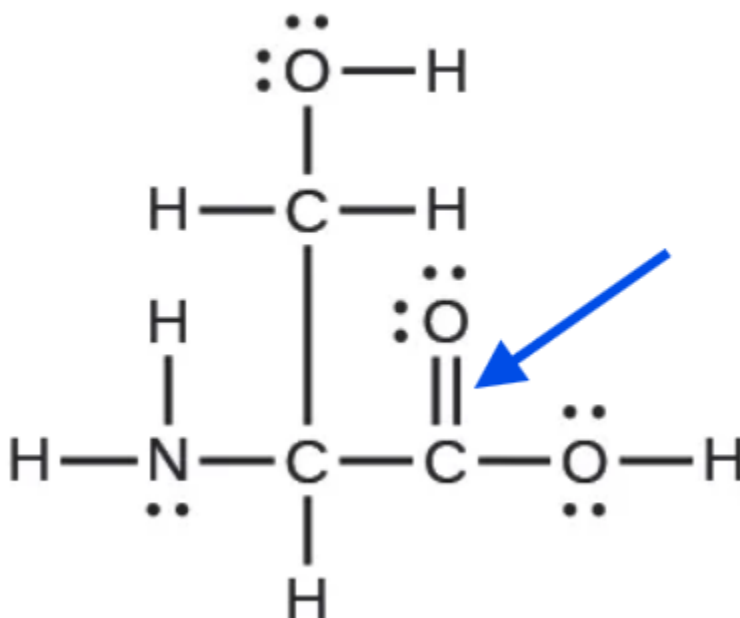
39. The arrangement of atoms in several biologically important molecules is given here. Complete the Lewis structures of these molecules by adding multiple bonds and lone pairs. Do not add any more atoms.

(a) the amino acid serine:



Exercise 7.39 (a) solution, unlabeled image alt text

39. (a)



The last three sentences of the alt text read (emphasis mine):

The oxygen atom is bonded to a hydrogen atom. The other carbon is **single bonded to two oxygen atoms, one of which is** bonded to a hydrogen atom. **The** oxygen atoms have two lone pairs of electron dots, and the nitrogen atom has one lone pair of electron dots.

One thing is wrong and two things could be slightly clarified:

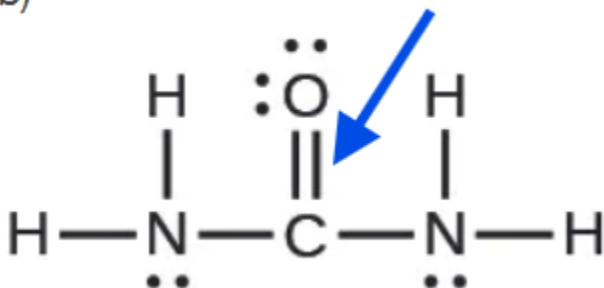
- The double bond in the image (see blue arrow above) is described as a single bond in the alt text.
- A couple of bonds with hydrogen are not labeled as single. Most bonds in the alt text specify the type, so it might not hurt to specify it for these bonds as well.
- Describing the lone pair electron dots at the end of the alt text is unusual compared to alt text for previous images. Perhaps specifying it applies to “all” the oxygen atoms would ensure the reader catches that?

Here is a proposed replacement (changes in bold):

The oxygen atom is **single** bonded to a hydrogen atom. The other carbon is **double bonded to one oxygen atom, and single bonded to another oxygen atom which is single** bonded to a hydrogen atom. **All three** oxygen atoms have two lone pairs of electron dots **each**, and the nitrogen atom has one lone pair of electron dots.

Exercise 7.39 (b) solution, unlabeled image alt text

(b)



Sentence 3 of the alt text reads (emphasis mine):

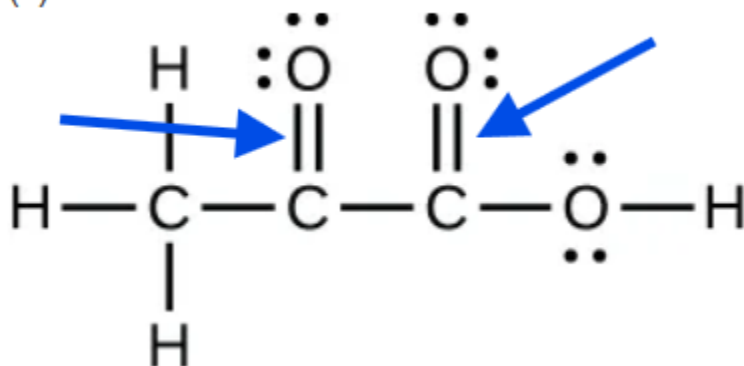
The carbon atom is **single** bonded to an oxygen atom and one nitrogen atom.

The carbon oxygen bond is double not single (blue arrow above). Suggest replacing sentence 3 with (changes in bold):

The carbon atom is **double** bonded to an oxygen atom and **single bonded to** one nitrogen atom.

Exercise 7.39 (c) solution, unlabeled image alt text

(c)



Sentences 3 and 4 of the alt text read:

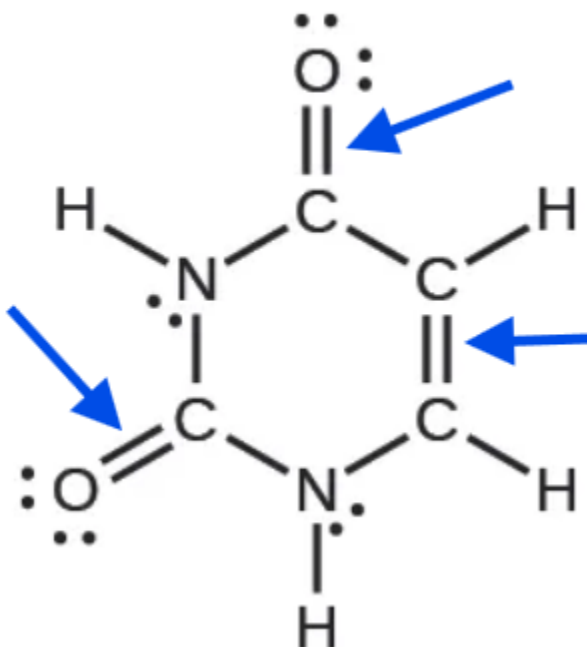
The carbon atom is single bonded to an oxygen atom and a third carbon atom. This carbon is then single bonded to two oxygen atoms, one of which is single bonded to a hydrogen atom.

The two double bonds (blue arrows above) are described as single in the alt text. I suggest replacing sentences 3 and 4 as follows (changes in bold):

The carbon atom is **double** bonded to an oxygen atom and **single bonded to** a third carbon atom. This carbon is then **double** bonded to **one oxygen atom and single bonded to another oxygen atom** which is single bonded to a hydrogen atom.

Exercise 7.39 (d) solution, unlabeled image alt text

(d)



Ok, this is getting old even for me. Anyway, the alt text reads:

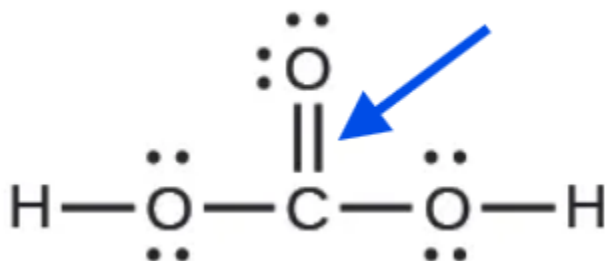
A Lewis hexagonal ring structure is shown. From the top of the ring, three carbon atoms, one nitrogen atom, a carbon atom and a nitrogen atom are single bonded to one another. The top carbon is single bonded to an oxygen, the second and third carbons and the nitrogen atom are each single bonded to a hydrogen atom. The next carbon is single bonded to an oxygen atom and the last nitrogen is single bonded to a hydrogen atom. The oxygen atoms have two lone pairs of electron dots, and the nitrogen atoms have one lone pair of electron dots.

Double bonds in the diagram are described as single bonds in the alt text (blue arrows above). Here is suggested replacement alt text (changes in bold):

A Lewis hexagonal ring structure is shown. From the top of the ring, three carbon atoms, one nitrogen atom, a carbon atom and a nitrogen atom **are bonded** to one another. **All of the ring bonds are single except for the second and third carbon, which are joined by a double bond.** The top carbon is **double** bonded to an oxygen, the second and third carbons and the nitrogen atom are each single bonded to a hydrogen atom. The next carbon is **double** bonded to an oxygen atom and the last nitrogen is single bonded to a hydrogen atom. The oxygen atoms have two lone pairs of electron dots, and the nitrogen atoms have one lone pair of electron dots.

Exercise 7.39 (e) solution, unlabeled image alt text

(e)



The alt text reads:

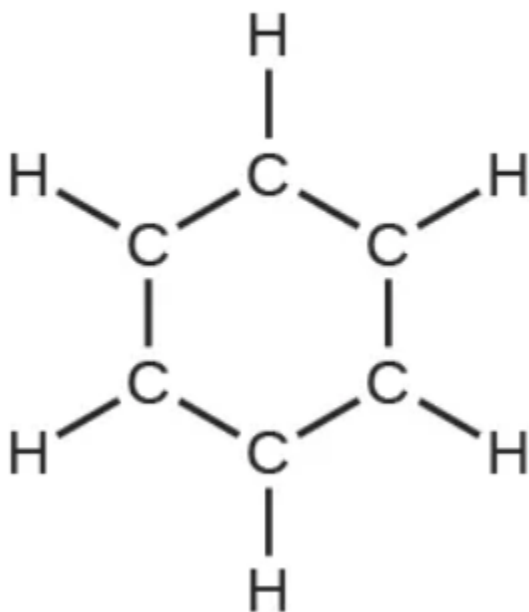
A Lewis structure is shown. A carbon atom is single bonded to three oxygen atoms. Two of those oxygen atoms are each single bonded to a hydrogen atom. Each oxygen atom has two lone pairs of electron dots.

A double bond (blue arrow above) is described as single in the alt text. Here is a suggested replacement alt text (changes in bold):

A Lewis structure is shown. A carbon atom is single bonded to **two** oxygen atoms, **which are** each single bonded to a hydrogen atom. **The carbon atom is also double bonded to a third oxygen atom.** Each oxygen atom has two lone pairs of electron dots.

Exercise 7.44 (d) unlabeled image in question, alt text

(d) benzene, C_6H_6 :



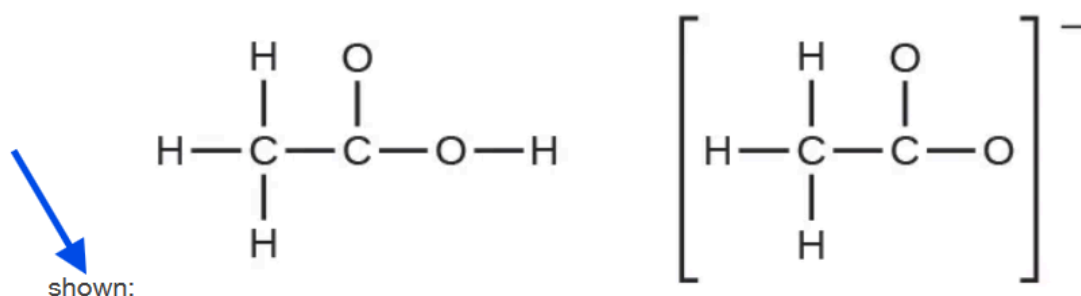
The alt text is (emphasis mine):

A Lewis structure shows a hexagonal ring composed of six carbon atoms. They form single bonds to **each another** and single bonds to one hydrogen atom each. I believe “each another” should be “**each other**”.

Exercises 7.48, 7.67, 7.73, and 7.74 question formatting

This may not matter, but there appears to be no paragraph break between the prose part of the 7.48 question and the image part. At some screen widths this can result in the word shown appearing orphaned (blue arrow in screenshot below). Don't most images in the book start a new paragraph or (in HTML terms) a new block level element?

48. In terms of the bonds present, explain why acetic acid, $\text{CH}_3\text{CO}_2\text{H}$, contains two distinct types of carbon-oxygen bonds, whereas the acetate ion, formed by loss of a hydrogen ion from acetic acid, only contains one type of carbon-oxygen bond. The skeleton structures of these species are



Exercise 7.67, 7.73, and 7.74 questions have the same formatting quirk.

Exercise 7.45 (a) solution, unlabeled image alt text

45. (a)

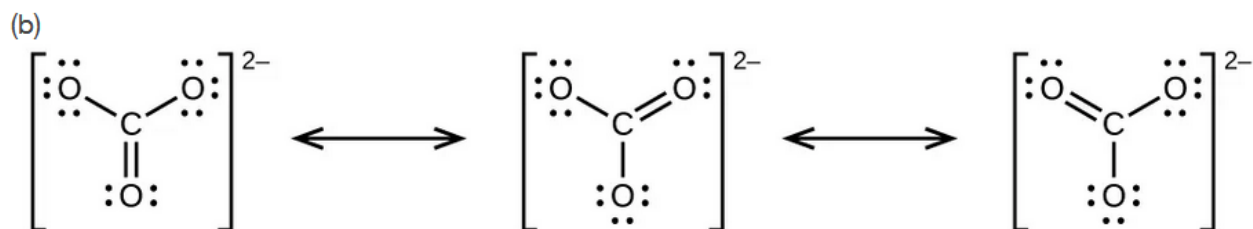


The last sentence of the alt text reads:

The sulfur atom in the right structures, also has a third electron dot below it.

This is not true, and this last sentence should be removed. Compare with alt text for exercise 8.15 (a) solution unlabeled image, which is the same image.

Exercise 7.45 (b) solution, unlabeled image alt text



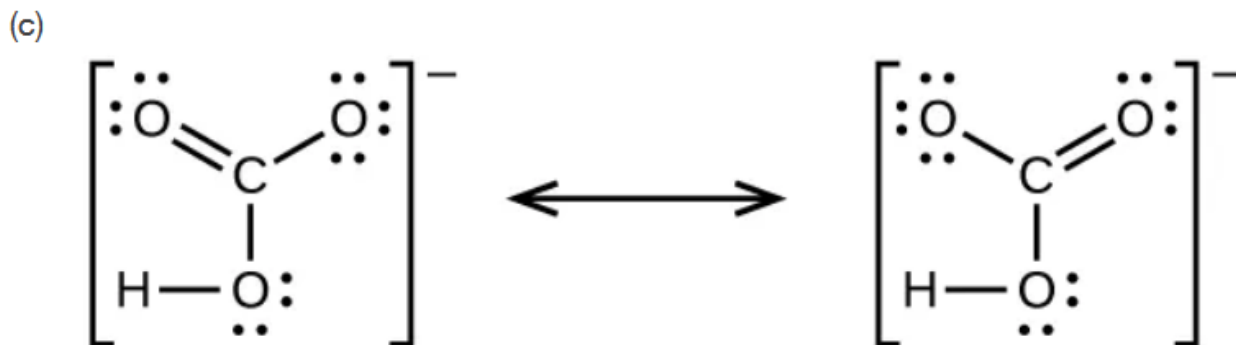
The 2nd to last sentence of the alt text reads (emphasis mine):

The central and right structures are the same as the first, but the position of the double bonded oxygen has moved to the **left oxygen in the right structure while the central structure only has single bonds.**

It is not true that the central structure has only single bonds. Here is a suggested replacement sentence (change in bold):

The central and right structures are the same as the first, but the position of the double bonded oxygen has moved to the **right oxygen in the center structure and to the left oxygen in the right structure.**

Exercise 7.45 (c) solution, unlabeled image alt text



The alt text reads (emphases mine):

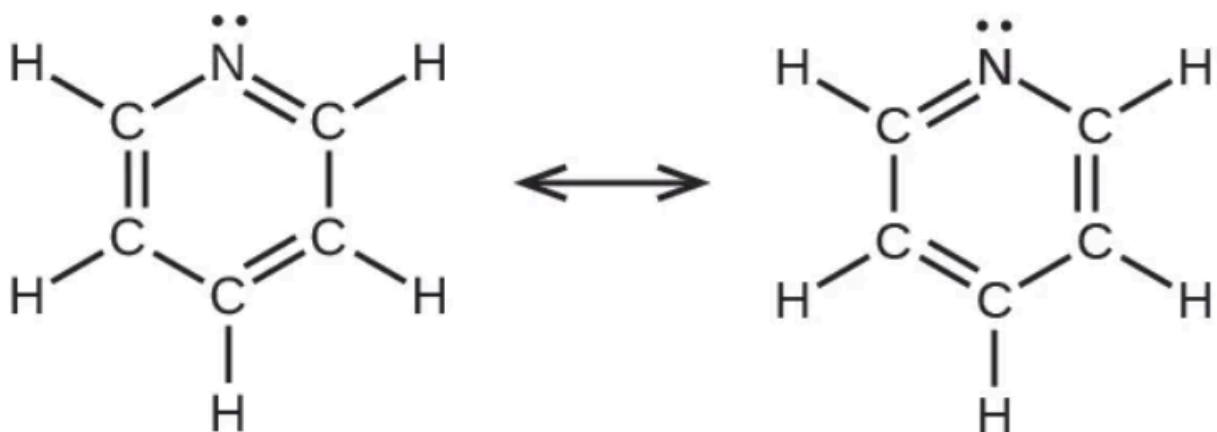
Two Lewis structures are shown, with a double-headed arrow in between, each surrounded by brackets and a superscripted negative sign. The left structure depicts a carbon atom bonded to three oxygen atoms. It is **single** bonded to one of these oxygen atoms, which has **three** lone pairs of electrons, and **double** bonded to the other two, **which have two lone pairs of electrons. One of the double bonded oxygen atoms also has** a single bond to a hydrogen atom. The right structure is the same as the first, but **there is only one double bonded oxygen. The oxygen with the single bonded hydrogen now has a single bond to the carbon atom.** The lone pairs of electrons have also changed to correspond with the bonds.

That alt text wrongly claims that the left structure has two oxygens with double bonds, and that mistake affects some other wording as well. Here is suggested replacement alt text (changes in bold):

Two Lewis structures are shown, with a double-headed arrow in between, each surrounded by brackets and a superscripted negative sign. The left structure depicts a carbon atom bonded to three oxygen atoms. It is **double** bonded to one of these oxygen atoms, which has **two** lone pairs of electrons, and **single** bonded to the other two. **One of the single bonded oxygen atoms has three lone pairs of electrons. The other single bonded oxygen atom has two lone pairs of electrons and a single bond to a hydrogen atom.** The right structure is the same as the first, but **the two oxygen atoms not bonded to the hydrogen atom have switched which one has the double bond to the carbon atom and which has the single bond to the carbon atom.** The lone pairs of electrons have also changed to correspond with the bonds.

Exercise 7.45 (d) solution, unlabeled image alt text

(d)



The final sentence of the alt text reads:

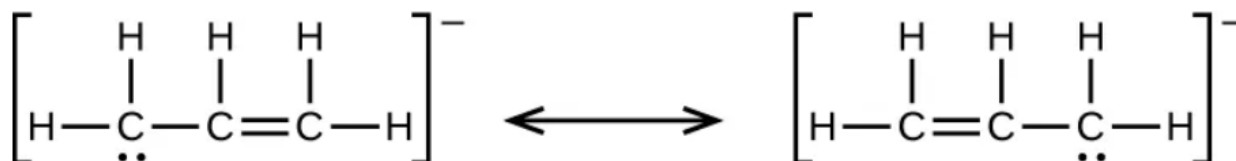
The right structure is the same as the first, but **each double bond has rotated to a new position.**

This is correct but seems a little unclear as to where the double bonds have “rotated” to. I suggest this (changes in bold):

The right structure is the same as the first, but **the alternating single and double bonds in the ring have switched, so what was a single bond on the left is a double bond on the right, and what was a double bond on the left is a single bond on the right.**

Exercise 7.45 (d) solution, unlabeled image alt text

(e)



The alt text reads:

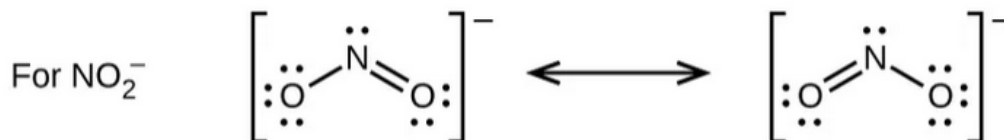
Two Lewis structures are shown with a double-headed arrow in between. The left structure shows a carbon atom single bonded to two hydrogen atoms and a second carbon atom. The second carbon atom is single bonded to a hydrogen atom and double bonded to a third carbon atom. The third carbon atom is single bonded to two hydrogen atoms. The whole structure is surrounded by brackets and a superscripted negative sign. The right structure shows a carbon atom single bonded to two hydrogen atoms and double bonded to a second carbon atom. The second carbon atom is single bonded to a hydrogen atom and a third carbon atom. The third carbon atom is single bonded to two hydrogen atoms. The whole structure is surrounded by brackets and a superscripted negative sign.

This omits the lone pair of electron dots on one of the carbons in each structure. Here is a suggested replacement alt text (changes in bold):

Two Lewis structures are shown with a double-headed arrow in between. The left structure shows a carbon atom single bonded to two hydrogen atoms and a second carbon atom **which has one lone pair of electron dots**. The second carbon atom is single bonded to a hydrogen atom and double bonded to a third carbon atom. The third carbon atom is single bonded to two hydrogen atoms. The whole structure is surrounded by brackets and a superscripted negative sign. The right structure shows a carbon atom single bonded to two hydrogen atoms and double bonded to a second carbon atom. The second carbon atom is single bonded to a hydrogen atom and a third carbon atom **which has one lone pair of electron dots**. The third carbon atom is single bonded to two hydrogen atoms. The whole structure is surrounded by brackets and a superscripted negative sign.

Exercise 7.47 solution, unlabeled image alt text

47.



The alt text reads (emphases mine):

A pair of Lewis structures are shown with a double-headed arrow in between the pair. The left structure **of the first pair** shows a nitrogen atom with one lone pair of electrons single bonded to an oxygen atom with three lone pairs of electrons. It is also double

bonded to an oxygen with two lone pairs of electrons. The right image **of this pair** depicts the mirror image of the left. Both images are surrounded by brackets and a superscripted negative sign. They are labeled, "For N O subscript two superscript negative sign."

Two things:

- There are a couple phrases suggesting this image has two pairs of resonance forms, when in fact it has one pair. I suggest removing those phrases (the ones in **bold red** above).
- The sentence describing that the right image is a mirror image of the left could be a bit clearer, especially if someone who cannot see the images conceives of the structure layouts and the plane of mirroring in a different way. I suggest a change in bold black below.)

Here is my suggested replacement alt text (removing the red bold phrases above as well as adding the change indicated in black bold below):

A pair of Lewis structures are shown with a double-headed arrow in between the pair. The left structure shows a nitrogen atom with one lone pair of electrons single bonded to an oxygen atom with three lone pairs of electrons. It is also double bonded to an oxygen with two lone pairs of electrons. The right image depicts the mirror image of the left, **switching which oxygen atom has two lone pairs of electrons and the double bond to the nitrogen atom**. Both images are surrounded by brackets and a superscripted negative sign. They are labeled, "For N O subscript two superscript negative sign."

Exercise 7.48 unlabelled image in question, alt text

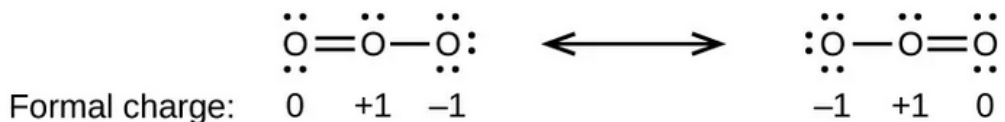
Sentence 1 of the alt text reads (emphasis mine):

Two Lewis structures are shown **with a double headed arrow in between**.

There is no arrow between the structures, and they are not described as resonance forms, so I don't think there should be. Suggest removing the phrase "with a double headed arrow in between".

Exercise 7.55 (a) solution, unlabeled image alt text

55. (a)



The alt text reads:

Two Lewis structures are shown with a double-headed arrow in between. The left structure shows an oxygen atom with one lone pair of electrons single bonded to an oxygen atom with three lone pairs of electrons. It is also double bonded to an oxygen atom with two lone pairs of electrons. The symbols and numbers below this structure read, "(0), (positive 1), (negative 1)." The phrase, "Formal charge," and a right-facing

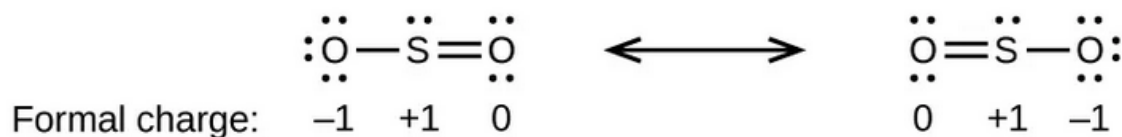
arrow lie to the left of this structure. The right structure appears as a mirror image of the left and the symbols and numbers below this structure read, “(negative 1), (positive 1), (0).”

This is correct except for mentioning a right facing arrow that is not present in addition to the double-headed arrow that really is present. However, presenting the atoms in the left structure in the order center atom, right atom, left atom, then later listing the formal charges in a different order, left atom, center atom, right atom, seems unnecessarily confusing. Here is a suggested replacement:

Two Lewis structures are shown with a double-headed arrow in between. Each atom in each structure has a number below it. At the left of these numbers is the label “Formal charge.” The left structure shows an oxygen atom with two lone pairs of electrons and the number 0 below it. That oxygen is double bonded to a second oxygen with one lone pair of electrons and the number positive 1 below it. The second oxygen is single bonded to a third oxygen atom with three lone pairs of electrons and the number negative 1 below it. The right structure appears as a mirror image of the left, with the double bonded oxygen with two lone pairs on the right side of the central atom instead of on the left side. The numbers below the right structure read negative 1, positive 1, 0, the opposite of the order of the numbers below the left structure, which were 0, positive 1, negative 1.

Exercise 7.55 (b) solution, unlabeled image alt text

(b)



The alt text reads:

Two Lewis structures are shown, with a double-headed arrow in between. The left structure shows a sulfur atom with one lone pair of electrons single bonded to an oxygen atom with three lone pairs of electrons. The sulfur atom also double bonded to an oxygen atom with two lone pairs of electrons. The symbols and numbers below this structure read, “(negative 1), (positive 1), (0).” The right structure appears as a mirror image of the left and the symbols and numbers below this structure read, “(0), (positive 1), (negative 1).”

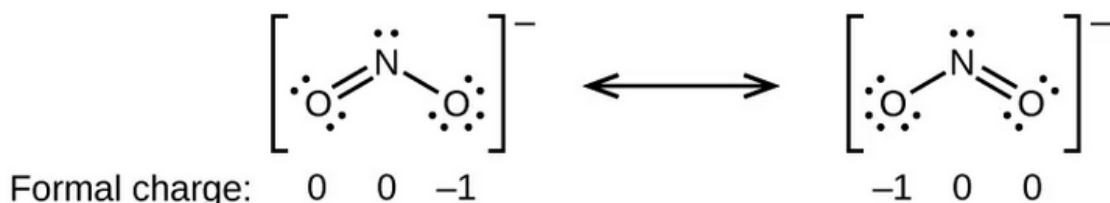
This is correct except for a minor grammatical error, a missing word “is” in sentence 3. However, like 7.55(a), presenting the atoms and the formal charges in different orders seems unnecessarily confusing. I suggest the following replacement alt text:

Two Lewis structures are shown, with a double-headed arrow in between. Each atom in each structure has a number below it. At the left of these numbers is the label “Formal charge.” The left structure shows an oxygen atom with three lone pairs of electrons and the number negative 1 below it. That oxygen is single bonded to a central sulfur atom with one lone pair of electrons and the number positive 1 below it. The sulfur atom is

also double bonded to an oxygen atom with two lone pairs of electrons and the number 0 below it. The right structure appears as a mirror image of the left, with the double bonded oxygen with two lone pairs on the left side of the central sulfur atom instead of on the right side. The numbers below the right structure read 0, positive 1, negative 1, the opposite of the order of the numbers below the left structure, which were negative 1, positive 1, 0.

Exercise 7.55 (c) solution, unlabeled image alt text

(c)



The alt text reads:

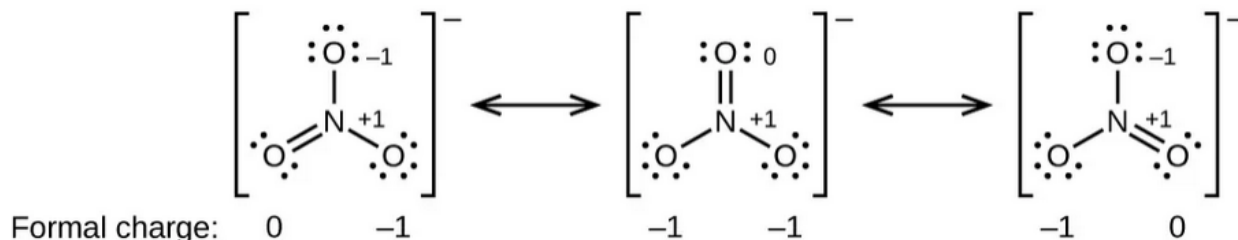
[Two Lewis structures are shown, with brackets surrounding each with a superscripted negative sign and a double ended arrow in between. The left structure shows a nitrogen atom with one lone pair of electrons single bonded to an oxygen atom with three lone pairs of electrons and double bonded to an oxygen atom with two lone pairs of electrons. The symbols and numbers below this structure read “open parenthesis, 0, close parenthesis, open parenthesis, 0, close parenthesis, open parenthesis, negative 1, close parenthesis. The right structure appears as a mirror image of the left and the symbols and numbers below this structure read “open parenthesis, negative 1, close parenthesis, open parenthesis, 0, close parenthesis, open parenthesis, 0, close parenthesis.]

This is correct. However, the bracket symbols surrounding the alt text are not needed (as opposed to the brackets described in words in the alt text). Also, what were parentheses symbols in the 7.55 (a) and (b) alt text are now unnecessary words describing “open parenthesis” and “close parenthesis” when no parentheses appear on the diagram. Finally, like 7.55 (a) and (b), , presenting the atoms and the formal charges in different orders seems unnecessarily confusing. I suggest the following replacement alt text:

Two Lewis structures are shown, with brackets surrounding each with a superscripted negative sign and a double ended arrow in between. Each atom in each structure has a number below it. At the left of these numbers is the label “Formal charge.” The left structure shows an oxygen atom with two lone pairs of electrons and the number 0 below it. That oxygen is bonded to a central nitrogen atom with one lone pair of electrons and the number 0 below it. The nitrogen atom is single bonded to an oxygen atom with three lone pairs of electrons and the number negative 1 below it. The right structure appears as a mirror image of the left, with the double bonded oxygen with two lone pairs on the right of the nitrogen atom instead of on the left. The numbers below the right structure read negative 1, 0, 0, the opposite of the order of the numbers below the left structure, which were 0, 0, negative 1.

Exercise 7.55 (d) solution, unlabeled image alt text

(d)



The alt text reads (surrounding square brackets part of alt text):

[Three Lewis structures are shown, with brackets surrounding each with a superscripted negative sign and a double ended arrow in between. The left structure shows a nitrogen atom single bonded to two oxygen atoms, each with three lone pairs of electrons and double bonded to an oxygen atom with two lone pairs of electrons. The single bonded oxygen atoms are labeled, from the top of the structure and going clockwise, “open parenthesis, negative 1, close parenthesis, open parenthesis, positive 1, close parenthesis”. The symbols and numbers below this structure read “open parenthesis, 0, close parenthesis, open parenthesis, negative 1, close parenthesis. The middle structure shows a nitrogen atom single bonded to two oxygen atoms, each with three lone pairs of electrons, one of which is labeled “open parenthesis, positive 1, close parenthesis” and double bonded to an oxygen atom with two lone pairs of electrons labeled “open parenthesis, 0, close parenthesis”. The symbols and numbers below this structure read “open parenthesis, negative 1, close parenthesis, open parenthesis, negative 1, close parenthesis. The right structure shows a nitrogen atom single bonded to two oxygen atoms, each with three lone pairs of electrons and double bonded to an oxygen atom with two lone pairs of electrons. One of the single bonded oxygen atoms is labeled, “open parenthesis, negative 1, close parenthesis while the double bonded oxygen is labeled, “open parenthesis, positive 1, close parenthesis”. The symbols and numbers below this structure read “open parenthesis, negative 1, close parenthesis” and “open parenthesis, 0, close parenthesis”.]

There are a few issues:

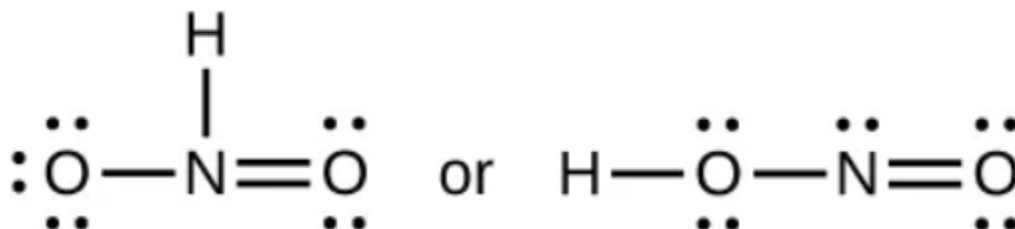
- The bracket symbols surrounding the entire alt text are not needed (as opposed to the brackets described in words in the alt text). Sentence 3 describes a single bonded oxygen atom labeled +1, which is wrong.
- What were parentheses symbols in the 7.55 (a) and (b) alt text are now unnecessary words describing “open parenthesis” and “close parenthesis” when no parentheses appear on the diagram.
- The numbers below the structures are described with different phrasing from the numbers beside atoms in the structures. This is confusing because they are the same thing, per-atom formal charge labels.
- All three structures are described in detail, unlike the alt text for 7.55 (a), (b), and (c), which just described additional structures as compared to the first.

Here is suggested replacement alt text:

Three Lewis structures are shown, with brackets surrounding each with a superscripted negative sign and a double ended arrow in between. Each atom is labeled with a number. The numbers are labeled as "Formal charge." The left structure shows a nitrogen atom single bonded to two oxygen atoms, each with three lone pairs of electrons and double bonded to an oxygen atom with two lone pairs of electrons. The single bonded oxygen atoms are labeled "negative 1," the double bonded oxygen atom is labeled "0", and the nitrogen atom is labeled "positive 1." The three oxygen atoms are arrayed around the central nitrogen atom with the double bonded oxygen atom to the lower left and the single bonded oxygen atoms to the lower right and at the top. The middle structure is similar but with the double bonded oxygen atom at the top. The right structure is similar but with the double bonded oxygen atom to the lower right. In all three structures, the double bonded oxygen atom is the one labeled "0" with two lone pairs of electrons.

Exercise 7.62 unlabelled image in question, alt text

62. Which of the following structures would we expect for nitro



The last sentence of the alt text reads (emphasis mine):

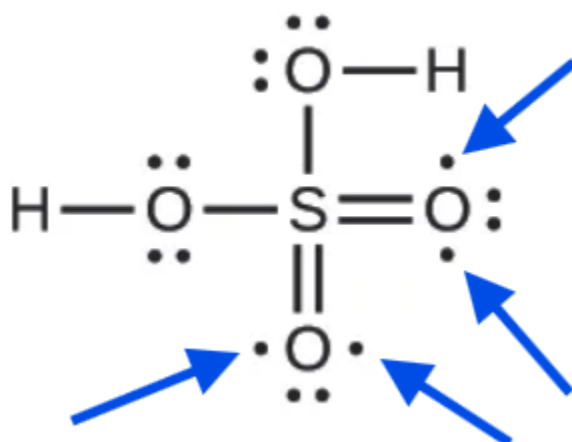
The oxygen atom is single bonded to a nitrogen atom **which is** double bonded to an oxygen atom with two lone pairs of electrons.

It does not mention the single lone pair of electrons on the nitrogen. I suggest replacing it with (changes in bold):

The oxygen atom is single bonded to a nitrogen atom **with one lone pair of electrons.**
The nitrogen atom is double bonded to an oxygen atom with two lone pairs of electrons.

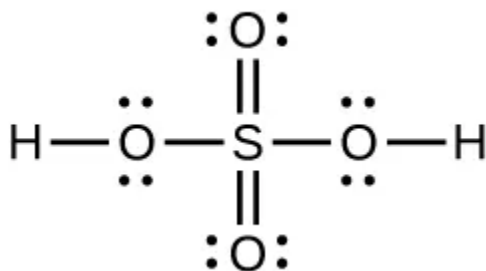
Exercise 7.63 solution, unpaired electrons

63.



I do not think the student has been given all of the information necessary to arrive at this Lewis structure at this point in the book. Three things confuse me.

- First, this Lewis structure for H₂SO₄ does not match the Lewis structure for the same molecule given in Chapter 8 in the Exercise 8.15 (d) solution, which does not appear to have unpaired electrons. I understand that some layout differences do not matter. However, the Exercise 7.63 solution has unpaired electrons (see blue arrows above), while the Exercise 8.15 (d) solution only has paired electrons (see screenshot below).



- Second, I do not think students have the knowledge to arrive at the unpaired electrons (see blue arrows) in the Exercise 7.63 solution Lewis structure. Chapter 7.3, section Writing Lewis Structures with the Octet Rule, advises students on creating Lewis structures by distributing lone pairs of electrons with no mention of distributing unpaired electrons. Chapter 7.3, section Exceptions to the Octet Rule, discusses unpaired electrons in molecules with an odd number of electrons, but this molecule has an even number of electrons and still has unpaired electrons.
- Third, the double bonds on two of the oxygen molecules make the sulfur atom hypervalent. Chapter 7.3, section Exceptions to the Octet Rule, discusses hypervalent molecules. However, the examples given in that section cover cases where the central molecule must be hypervalent due to the number of other atoms that must be bonded to (like SF₆) or due to the number of valence electrons that must be distributed (like XeF₄). That section does not contain an example where a central atom is made hypervalent by doubling bonds. Would it make sense to add such an example, either to Chapter 7.3 or

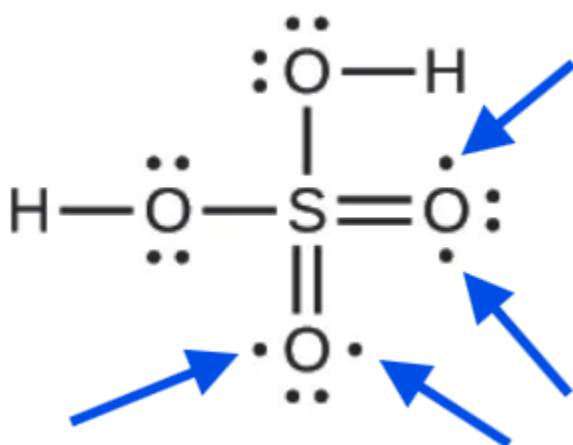
to Chapter 7.4 where formal charges are discussed? (I mention Chapter 7.4 because I assume the heuristic that results in the double bonds is based on minimizing formal charges.)

Note: On this third source of confusion, see also my Chapter 8 errata document, specifically the errata on Exercise 8.15 (c) and (d) solutions, unlabeled figures.

Exercise 7.63 solution, unlabeled image alt text

Note: If the image is changed at all in response to the previous errata, this errata may change or go away.

63.



The alt text reads:

A Lewis structure shows a hydrogen atom single bonded to an oxygen atom with two lone pairs of electrons. The oxygen atom is single bonded to a sulfur atom. The sulfur atom is double bonded to two oxygen atoms, each of which have three lone pairs of electrons, and single bonded to an oxygen atom with two lone pairs of electrons. This oxygen atom is single bonded to a hydrogen atom.

Sentence 3 describes the double bonded oxygen atoms as each having “three lone pairs of electrons.” However, the diagram only shows 4 electrons around each of those oxygen atoms, not 6. Two the electrons on each double bonded oxygen are in a lone pair, the remaining two are alone and not paired up (see blue arrows above).

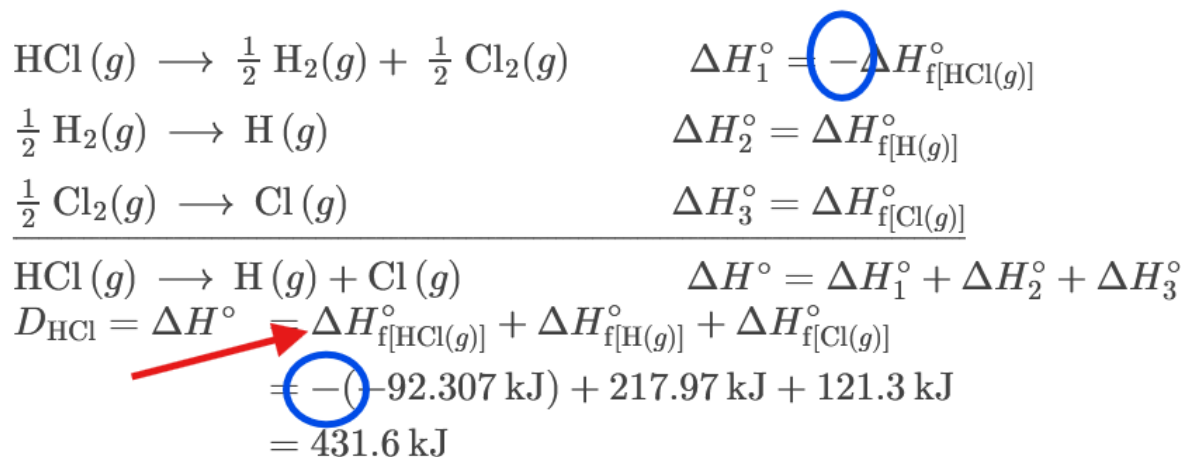
Here is a suggested replacement alt text (changes in bold):

A Lewis structure shows a hydrogen atom single bonded to an oxygen atom with two lone pairs of electrons. The oxygen atom is single bonded to a sulfur atom. The sulfur atom is double bonded to two oxygen atoms. **Each double bonded oxygen atom has four electron dots, two together as a lone pair, and two separated on opposite sides of the oxygen symbol, “O”.** The sulfur atom is also single bonded to an oxygen atom with two lone pairs of electrons. This oxygen atom is single bonded to a hydrogen atom.

Exercise 7.69 solution

While the numerical answer to 7.69 in the book is correct, 431.6 kJ, I believe there is an error, a dropped negative sign, in one line of the equations leading up to the final answer. Specifically, notice how $\Delta H^\circ_1 = -\Delta H^\circ_{f[\text{HCl}(g)]}$ in the top line (first blue circle on screenshot below). Then from the first line after the horizontal rule to the second line after the horizontal rule, ΔH°_1 becomes $\Delta H^\circ_{f[\text{HCl}(g)]}$ without the negative sign (red arrow). By the third line after the horizontal rule, the negative sign is back in $-(-92.307 \text{ kJ})$, where -92.307 kJ is the $\Delta H^\circ_{f[\text{HCl}(g)]}$ value from Appendix G.

69.



Exercise 7.73 question phrasing

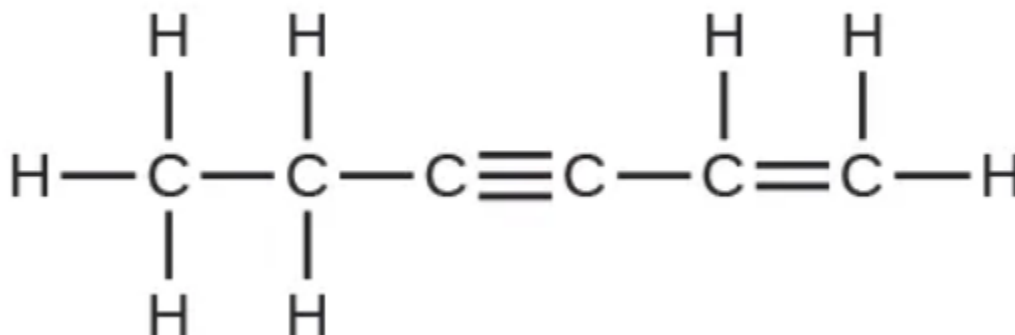
The text part of the question reads:

Complete the following Lewis structure by adding bonds (not atoms), and then indicate the longest bond:

It is then followed by a molecular structure without the bonds filled in. The way the sentence is phrased, it is easy for a student to assume that the “longest bond” part of the question is looking for one particular bond in the molecular structure, not one type of bond, the single carbon-carbon bonds on the structure. Since the first part of the question asks the student to make a Lewis structure, the student could assume they are supposed to circle a single longest bond. Would it be clearer to replace “the longest bond:” with “the longest **bond or bonds**:” I would suggest “the longest bond types:” but then the student might assume just “single bonds” is a good enough answer.

Exercise 7.73 solution, unlabeled image alt text

73.



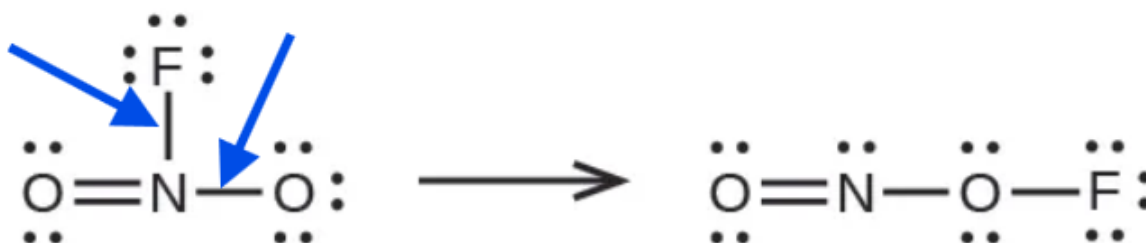
Sentence 4 of the alt text reads (emphasis mine):

The second carbon atom is single bonded to a third carbon atom that is triple bonded to a fourth carbon atom **and** single bonded to a fifth carbon atom.

The word “and” in sentence 4 makes it sound like the third carbon atom is both triple bonded to the fourth carbon atom and single bonded to the fifth, when in fact it is the fourth carbon atom single bonded to the fifth. I suggest replacing the word “and” with “**which is**”.

Exercise 7.74 unlabeled figure in question, alt text

74. Use the bond energy to calculate an approximate value of ΔH for the following reaction: the more stable form of FNO_2 ?



Sentence 3 reads (emphasis mine)

It is also **bonded** to a fluorine atom and another oxygen atom, each with three lone pairs of electrons.

It might be slightly clearer if “bonded” was replaced with “**single bonded**”. All other bonds in this alt text are explicitly specified as single or double. The bonds mentioned are shown with blue arrows in the screenshot above.

Exercise 7.93 (e) solution

The 7.93 (e) solution reads (emphasis mine):

(e) electron-pair geometry: trigonal **bypyramidal**

I believe “bypyramidal” is supposed to be spelled “**bipyramidal**”, with the second letter being an “i” not a “y”. Compare, for example, with the spelling in Exercise 7.95 (f) solution, Figure 7.16, and Figure 7.19

Exercise 7.97 (f) solution and Chapter 7.2 and 7.6 polar bond definition ambiguity

The 7.97 question asks:

Which of the following molecules and ions contain polar bonds? Which of these molecules and ions have dipole moments?

7.97 (f) specifically asks that question about PH_2^- . The 7.97 solution says all (including f) contain polar bonds, and that PH_2^- is one that has a dipole moment. This makes sense if any non-zero electronegativity difference creates polar bonds, since the ends of the bonds are not identical.

The confusing part is that Chapter 7.2 says that bonds with a very small electronegativity difference are nonpolar. Specifically, Chapter 7.2, section Electronegativity and Bond Type, paragraph 1, sentence 2 says (emphases mine):

When the difference is **very small** or zero, the bond is covalent and **nonpolar**.

The word “difference” here refers to an electronegativity difference. Chapter 7.2, figure 7.6 shows H and P both having electronegativities of 2.1, meaning their electronegativity difference would be (near) 0. Chapter 7.2, figure 7.8 has a table in the lower right that implies a rough cutoff point in bond type around a 0.4 electronegativity difference (screenshot below). Granted, the text immediately after figure 7.8 says it is a “rough approximation” and “just a general guide”.

Bond Type	Electronegativity Difference
pure covalent	< 0.4
polar covalent	between 0.4 and 1.8
ionic	> 1.8

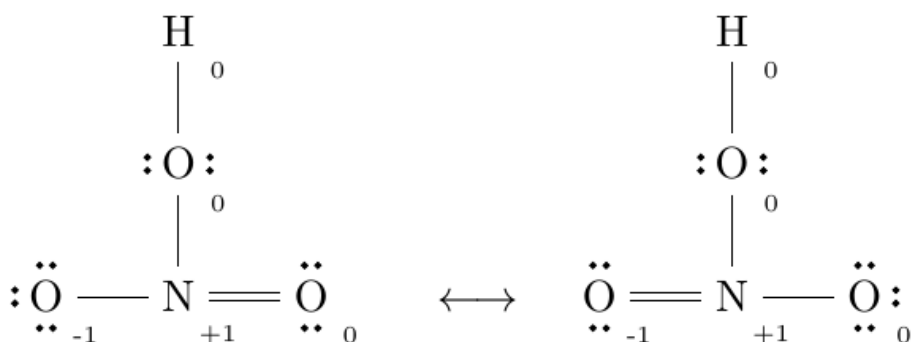
In summary, I think the book answer is exercise 7.97 (f) is probably correct, but the way bond types are described in Chapter 7.2 makes it seem at least ambiguous. Perhaps this could be clarified somewhere?

Exercise 7.105 (c), (e), and (f) solution consistency

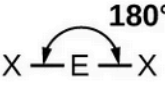
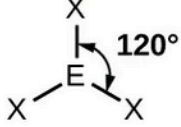
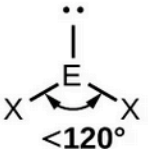
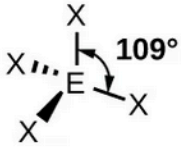
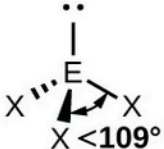
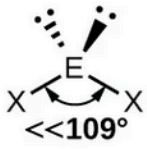
Why does the solution “(e) bent ($<120^\circ$)” have a less than sign but the (c) and (f) solutions of “bent (109°)” do not have a less than sign? It seems inconsistent.

Exercise 7.105 (e) solution

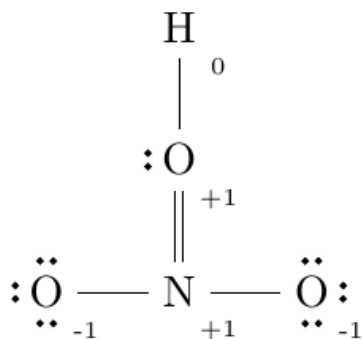
Could someone please double check the book solution for Exercise 7.105 (e)? The book gives the solution as “(e) bent ($<120^\circ$)”. However, I think the actual solution might be “(e) bent ($<109^\circ$)”. My reasoning is as follows. From my work and [online resources](#), I think the Lewis structure for nitric acid is like this (labels are formal charges):



If correct, this means that the oxygen in the OH group has four regions of electron density and two lone pairs. The Chapter 7.6, Figure 7.19 cell for four electron regions and two lone pairs gives a molecular structure shape as bent ($<109^\circ$), which makes sense given the tetrahedral electron-pair geometry.

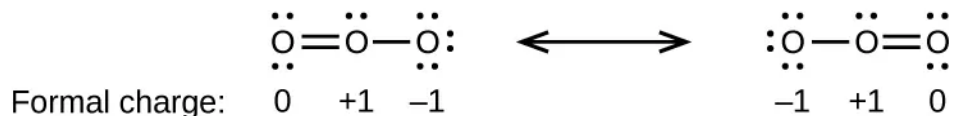
Number of electron regions	Electron region geometries: 0 lone pair	1 lone pair	2 lone pairs
2	 Linear		
3	 Trigonal planar	 Bent or angular	
4	 Tetrahedral	 Trigonal pyramid	 Bent or angular

An alternate Lewis structure would give the book answer of bent ($<120^\circ$), but that Lewis structure has more non-zero formal charges, as shown below, so I do not think that structure would be preferred.

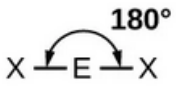
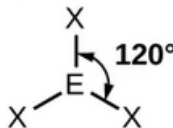
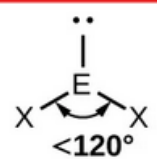


Exercise 105 (f) solution

Could someone please double check the book solution for Exercise 7.105 (f)? The book gives the solution as "(f) bent (109°)". However, I think the actual solution might be "(f) bent ($<120^\circ$)". My reasoning is as follows. The Exercise 55 (a) solution gives the Lewis structure for an ozone (O_3) molecule as follows.



Looking at the central oxygen atom shows three regions of electron density with one lone pair. The Chapter 7.6, Figure 7.19 cell for 3 electron regions but 1 lone pair gives the molecular structure as bent ($<120^\circ$).

Number of electron regions	Electron region geometries: 0 lone pair	1 lone pair
2	 Linear	
3	 Trigonal planar	 Bent or angular

Exercise 7.111 question

The 7.111 question reads (emphasis mine):

A compound with a molar mass of about 42 g/mol contains 85.7% carbon and 14.3% hydrogen. **What is its molecular structure?**

The solution shows a molecular structure diagram. It might be clearer to say “**Draw its molecular structure.**” instead of “What is its molecular structure?” similar to the way the question is worded for Exercise 7.109. I would hate for a student to try explaining it with words.

Exercises 7.112 through 7.116

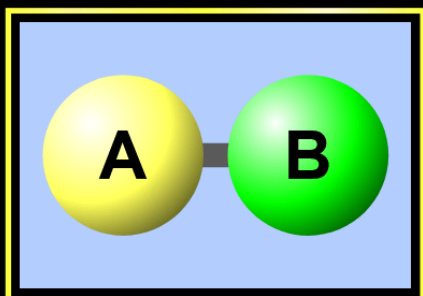
See chapter-wide section on Molecular Simulator Accessibility

Exercise 7.113

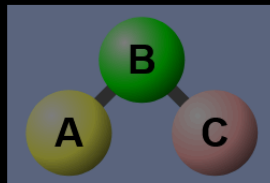
The question involves a linked simulation:

<https://phet.colorado.edu/en/simulations/molecule-polarity>. The question asks you to use the simulation for a “real molecule”. There is a button for “Real molecules.”

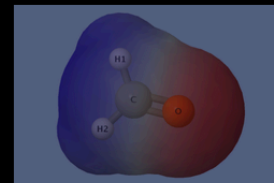
Molecule Polarity



Two Atoms



Three Atoms



Real Molecules

However, when you click that button, you just get a message saying:

Real Molecules is not currently available in HTML5.

If your device runs Java, you can use the Java version at

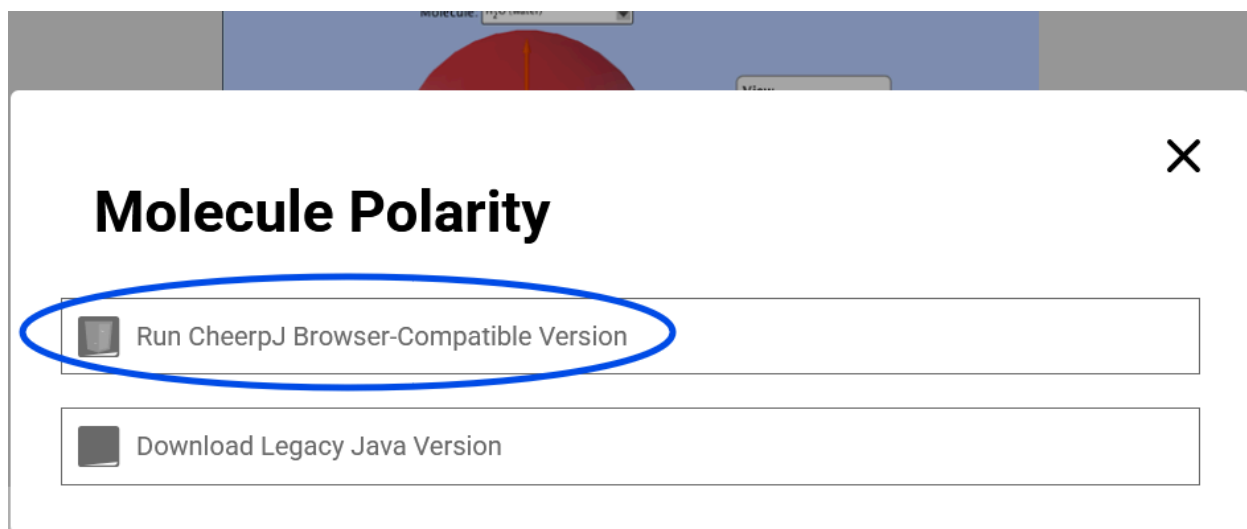
<https://phet.colorado.edu/en/simulations/legacy/molecule-polarity/translations>

Real Molecules is not currently available in HTML5.

If your device runs Java, you can use the Java version at

<https://phet.colorado.edu/en/simulations/legacy/molecule-polarity/translations>

In addition to being the kind of message that makes a user think something is broken or they did something wrong, even a savvy user may know they do not have Java installed on their computer, so may give up. This is unfortunate, because if you click on the link, and click the play button on the resulting page, you get a button offering a version that **does** work without Java installed, “Run CheerpJ Browser-Compatible Version.”



Clicking on that button takes you to a working simulation (at least in Firefox on Windows 11).

It would be nice if this whole interaction was cleaned up, but I know time and resources are limited. However, it would help a little just to change the message:

Real Molecules is not currently available in HTML5.

If your device runs Java, you can use the Java version at

<https://phet.colorado.edu/en/simulations/legacy/molecule-polarity/translations>

Changing it to something like this would help:

Real Molecules is not currently available in HTML5.

You can run an older version at

<https://phet.colorado.edu/en/simulations/legacy/molecule-polarity/translations>

Of course, if that same message appears for other simulations, someone would need to make sure the modified statement was true for those simulations as well. I guess there is no free lunch.

It might be better just to link directly to the old version in Exercise 7.113. However, I understand that could be confusing too because you would still want Exercise 7.112 (and some examples in the chapter maybe) to link to the new version because they use parts of the simulation other than the "Real molecules" section, parts which already work in HTML5.

Exercise 7.113 solution is incomplete

The 7.113 question has three parts, (a) through (c). However, the 7.113 solution only contains:

The molecular dipole points away from the hydrogen atoms.

This only answers part (c), and not very well. There are no answers at all given for parts (a) and (b).